MAHARASHTRA STATE ELECTRICITY DISTRIBUTION CO. LTD.

MSEDCL | Powering Maharashtra

PROJECT EXECUTION BLUEPRINT

Electric Pole Geo-Mapping & AI-Powered Infrastructure Intelligence System

Full Maharashtra — All 36 Districts

T0: April 1, 2026 → T6: September 30, 2026

Document Version	Project Status	T0 (Start)	T6 (End)	Classification
V1.0 — 2026	ACTIVE BLUEPRINT	April 1, 2026	September 30, 2026	CONFIDENTIAL

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Chapter 1: Executive Summary

1.1 Project Overview

Maharashtra State Electricity Distribution Co. Ltd. (MSEDCL) operates one of India's largest electricity distribution networks, serving over 2.9 crore consumers across all 36 districts of Maharashtra. The backbone of this network — millions of electric poles — currently exists without any centralized, GPS-tagged, digitally accessible asset record.

This blueprint presents the complete execution plan for the MSEDCL Electric Pole Geo-Mapping and AI-Powered Infrastructure Intelligence System — a technology-driven initiative to survey, map, classify, and manage every electric pole in Maharashtra within a single, aggressive 6-month window.

⚠ CRITICAL PROJECT PARAMETER

T0 = April 1, 2026 (Project Start — Day Zero)
T6 = September 30, 2026 (Full Completion & Handover)
Total Duration = 6 Months (183 days)
This is a high-intensity, compressed-timeline deployment. Every week counts.

1.2 Project Vision

★ To create a first-of-its-kind, AI-powered, GIS-enabled, real-time electric pole asset registry
covering ALL 36 districts of Maharashtra — fully surveyed, classified, and handed over to MSEDCL within 6 months from April 1, 2026.

1.3 Core Problem Being Solved

Problem	Impact on MSEDCL	Our Solution
No GPS coordinates for poles	Cannot locate assets on any map	Mobile + Drone geo-tagging
No centralized pole database	Data fragmented across 1,000+ sub-divisions	Cloud-based central repository
Manual inspection only	Slow, error-prone, dangerous	AI visual classification from photos
No condition monitoring	Purely reactive maintenance	AI condition scoring (1–5 scale)
No asset lifecycle tracking	Unknown pole age & full history	Unique Pole ID + digital history
High operational cost	Inefficient unplanned manpower use	Predictive AI maintenance engine

1.4 6-Month Execution Summary

Month	Period	Phase Name	Key Focus
Month 1	Apr 1 – Apr 30	Mobilization & Pilot	Hiring, training, system setup, 2-week pilot in 1 district
Month 2	May 1 – May 31	Full Deployment — Wave 1	All 36 districts activated; 300 surveyors in field
Month 3	Jun 1 – Jun 30	Intensive Survey + AI Go-Live	Peak survey operations; AI system deployed
Month 4	Jul 1 – Jul 31	Continuous Survey + QA	Data quality drive; AI model refinement; 60% target
Month 5	Aug 1 – Aug 31	Survey Completion + Integration	Complete remaining poles; GIS dashboard live
Month 6	Sep 1 – Sep 30	Validation, AI Handover & Closeout	Final validation, AI dashboard, handover to MSEDCL

1.5 Key Deliverables — September 30, 2026

- Complete GIS map of all electric poles across all 36 districts of Maharashtra
- Central cloud database — every pole with full attributes, photos, and Unique Pole ID
- AI-powered pole classification system (type + condition) — operational and live
- AI Dashboard — MSEDCL management interface with full pole map, analytics, and features
- Final comprehensive project report submitted to MSEDCL

1.6 Project Scale

Parameter	Value
Project Duration	6 Months (April 1 – September 30, 2026)
Districts Covered	All 36 districts of Maharashtra — simultaneously
Estimated Total Poles	55–75 Lakh (refined after pilot)
Pilot Phase	2–3 weeks in 1 district (April 1–21, 2026)
Full Deployment Start	April 22, 2026 (all 36 districts simultaneously)
Field Surveyors	~300 (hired and trained within Month 1)
Field Supervisors	~60
Technology Stack	Mobile App + Cloud + AI/ML + GIS + Drone
AI System Live	By end of Month 3 (June 30, 2026) + 15 buffer days
Final Handover	September 30, 2026

1.7 Three Systems Combined — The Strategic Advantage

► FIELD SYSTEM → 300 Surveyors + Mobile App + Drones = Real-Time Ground-Level Data
► DATA SYSTEM → Cloud + PostgreSQL/PostGIS + APIs = Centralized Scalable Infrastructure
► INTELLIGENCE SYSTEM → AI Models + Automation + AI Dashboard = Smart Decision Making

This is NOT a survey project. It is a SMART INFRASTRUCTURE INTELLIGENCE SYSTEM that will serve MSEDCL for the next 20+ years.

Chapter 2: Introduction

2.1 About MSEDCL

Parameter	Details
Established	2005 (following MSEB restructuring under Electricity Act, 2003)
Headquarters	Mumbai, Maharashtra
Service Area	All 36 districts of Maharashtra (excl. Mumbai city)
Consumer Base	Approx. 2.9 Crore consumers
Employee Strength	~80,000+
Circles / Divisions	15 Circles, 50+ Divisions, 1,000+ Sub-Divisions
Annual Revenue	Approx. ₹70,000 Crore+
Network	Lakhs of KM of LT/HT/EHT lines across 36 districts

2.2 Background and Context

Electric distribution poles form the critical last-mile infrastructure that connects power generation to every consumer — every lit home, every running factory, every hospital. Despite their absolute operational criticality, MSEDCL currently has no comprehensive, geo-tagged, digitally accessible inventory of its pole network across Maharashtra.

As of April 2026, pole records are maintained in paper registers at sub-division level, no GPS coordinates are linked to individual poles, condition assessment happens only during reactive maintenance, and there is no centralized visibility for planning, compliance, or infrastructure investment decisions.

As Maharashtra's energy demand rises year on year, infrastructure failures caused by aged, damaged, or untracked poles result in consumer outages, expensive emergency repairs, and avoidable accidents. The need for technology-first asset management has reached a point of urgency that this project is directly designed to address.

2.3 The Digital Infrastructure Gap

★ KEY INSIGHT: In April 2026, MSEDCL can accurately tell you how many units of electricity were
consumed in a village last month — but cannot tell you how many poles stand in that village,
what condition they are in, when they were last inspected, or where exactly they are on a map.
THIS is the gap that this 6-month project will permanently close.

2.4 Why a 6-Month Compressed Timeline?

- MSEDCL's strategic modernization roadmap demands immediate action — not a multi-year program
- Mobile + AI + GIS technology in 2026 enables high-speed, high-quality field data collection at scale
- Parallel deployment across all 36 districts simultaneously — enabled by the 300-surveyor field force — makes 6 months achievable
- Regulatory and compliance pressures from CEA and CERC for digital asset records are time-sensitive
- Smart Grid and Smart Metering initiatives require a foundational pole asset layer — urgently

2.5 Scope of This Blueprint

This document is the definitive Project Execution Blueprint for the MSEDCL Pole Mapping & AI System. It covers every aspect of project delivery — field operations, technology architecture, AI development, manpower planning, budget, permissions, risk management, SOPs, and final deliverables. All timelines are anchored to T0 = April 1, 2026.

Chapter 3: Objectives

3.1 Primary Objectives

1. Conduct a complete geo-referenced survey of every electric pole across all 36 districts of Maharashtra by September 30, 2026.
2. Assign a permanent Unique Pole ID (UID) to every pole surveyed — creating a digital identity for each asset for its entire operational lifecycle.
3. Capture 30+ attributes per pole including GPS coordinates, pole type, material, height, circuit type, condition score, equipment attached, and ownership zone.
4. Build a centralized, cloud-hosted pole asset database accessible in real-time to all authorized MSEDCL circles and divisions.
5. Deploy an AI-powered classification system for automated pole type recognition and condition assessment — live by end of Month 3 (June 30, 2026).
6. Deliver a fully functional AI Dashboard — MSEDCL's primary interface for viewing the pole map, analytics, alerts, and maintenance intelligence.
7. Integrate drone-based aerial mapping for forest, tribal, and difficult-terrain areas where ground survey is constrained.
8. Complete formal handover of all systems, data, documentation, and trained MSEDCL staff by September 30, 2026.

3.2 Secondary Objectives

- Enable predictive maintenance by identifying high-risk poles before failure occurs
- Reduce future field inspection costs through AI-assisted remote condition monitoring
- Provide district-wise and division-wise asset inventory reports for MSEDCL annual planning
- Build a digital maintenance history log per pole for regulatory compliance and audit trails
- Serve as the foundational asset layer for MSEDCL's long-term Smart Grid initiative

3.3 SMART Objectives Framework

Objective	Specific	Measurable	Achievable	Relevant	Time-bound
Pole Survey	All 36 districts	Total pole count	300 surveyors deployed	Core asset data	By Sep 30, 2026
Unique ID	Every pole	100% UID assigned	Mobile app auto-gen	Lifecycle tracking	During survey
AI Classification	Type + Condition	≥90% accuracy	Transfer learning ML	Ops & safety	By Jun 30, 2026
AI Dashboard	Live MSEDCL interface	All features live	Cloud + GIS platform	Management ops	By Aug 31, 2026
Central Database	All pole attributes	100% data stored	Cloud redundancy	Planning & audit	By Sep 30, 2026

3.4 Key Performance Indicators (KPIs)

KPI	Target	Measurement Method	Review Frequency
Survey Completion Rate	≥ 98% of all poles	DB count vs. estimate	Weekly
Daily Survey Productivity	≥ 50 poles / surveyor / day	App-based daily tracking	Daily
GPS Accuracy	± 3 meters or better	Drone cross-validation	Weekly sampling
Data Completeness	≥ 95% fields filled / record	Automated QC check	Daily
AI Model Accuracy — Type	≥ 90% on validation set	Hold-out test dataset	Monthly
AI Model Accuracy — Condition	≥ 85% on validation set	Ground-truth comparison	Monthly
Dashboard Uptime	≥ 99.5% availability	Cloud monitoring alerts	Continuous
Pilot Completion	100% pilot district in 3 weeks	Milestone tracker	End of Week 3
AI System Live	Deployed by Jun 30, 2026	Production deployment log	End of Month 3
Final Handover	100% by Sep 30, 2026	Handover checklist sign-off	End of Month 6

Chapter 4: Study Area — Ground Reality

4.1 Maharashtra State — Geographic Profile

Parameter	Data
Total Area	3,07,713 sq. km (3rd largest state in India)
Total Districts	36 (ALL to be covered in 6 months)
MSEDCL Circles	15 Circles
MSEDCL Divisions	~50 Divisions
MSEDCL Sub-Divisions	~1,000+ Sub-Divisions
Agro-climatic Zones	5 — Vidarbha, Marathwada, Konkan, Western Maharashtra, North Maharashtra
Total Villages	~44,000
Urban Local Bodies	~534
Topography	Coastal plains, Western Ghats, Deccan plateau, river valleys, semi-arid zones

4.2 MSEDCL Circle-wise Coverage Map

Circle	Key Districts	Terrain Challenge	Survey Priority
Pune	Pune, Satara (part)	Dense urban + peri-urban	High — urban complexity
Nashik	Nashik, Dhule, Nandurbar	Semi-arid + hilly + tribal	High — pilot candidate
Aurangabad	Aurangabad, Jalna, Beed	Marathwada dry, remote villages	High
Nagpur	Nagpur, Wardha	Urban + agricultural mix	High
Amravati	Amravati, Akola, Washim	Vidarbha heat, some forest	Medium-High
Latur	Latur, Osmanabad, Nanded	Arid, sparsely connected	Medium
Konkan	Ratnagiri, Sindhudurg	Coastal, dense forest, access issues	Medium — drone needed
Kolhapur	Kolhapur, Sangli, Solapur	Flat agricultural, accessible	Medium
Thane	Thane, Palghar, Raigad	Urban sprawl + tribal forest	High — drone needed
Chandrapur	Chandrapur, Gadchiroli	Dense forest, tribal, LWE sensitivity	High — drone + permissions
Other Circles (5)	Remaining districts	Mixed terrain	Standard

4.3 Terrain Classification — Survey Approach per Zone

Terrain Type	% of State	Representative Districts	Primary Survey Method
Urban / Peri-urban	~15%	Pune, Nagpur, Nashik, Thane, Aurangabad	Mobile app — dense data, vehicle not always needed
Agricultural Flat Plains	~45%	Vidarbha, Marathwada, W. Maharashtra	Mobile app + motorcycle — high productivity zone
Semi-arid / Rocky Terrain	~15%	Marathwada, N. Maharashtra	Mobile app + 4WD SUV for team transport
Hilly / Western Ghats	~10%	Konkan, Kolhapur, E. Nashik	Mobile + drone for ridge-line areas
Dense Forest / Tribal	~10%	Gadchiroli, Chandrapur, Palghar, Nandurbar	Drone primary + limited ground team
Coastal / Wetland	~5%	Ratnagiri, Sindhudurg, Raigad	Mobile + drone + boat (select areas)

4.4 Estimated Pole Distribution

Pole Category	Estimated Count	Basis
LT Distribution Poles (last-mile)	35–45 Lakh	~1 pole per 5–8 LT consumers
HT Line Poles / Structures	8–12 Lakh	HT network route length estimates
11 kV Feeder Poles	5–8 Lakh	Feeder route lengths from MSEDCL
Service Connection Support Poles	2–4 Lakh	Dense urban area estimates
Pole-Mounted Transformers	1.5–2.5 Lakh	MSEDCL transformer count data
MSEDCL-Owned Street Light Poles	1–2 Lakh	Municipal distribution data
TOTAL ESTIMATED POLES	55–75 Lakh	Subject to pilot validation — April 2026

4.5 Pilot District — Selection & Rationale

The 2–3 week pilot (April 1–21, 2026) will be conducted in one district before the full 36-district simultaneous rollout begins on April 22, 2026.

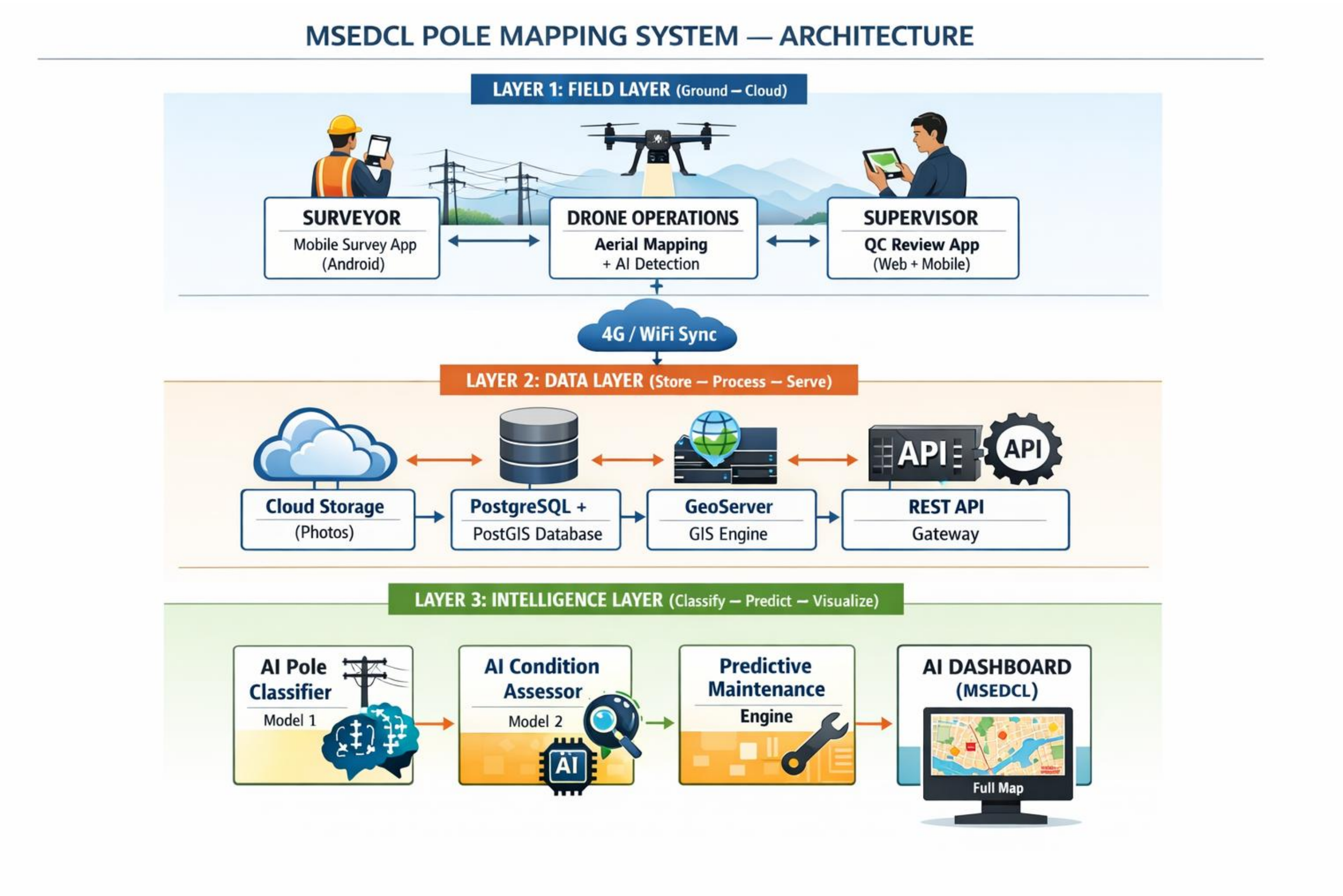
Selection Criterion	Requirement
Terrain mix	Must include urban, semi-urban, and rural poles for comprehensive testing
MSEDCL cooperation	Active divisional office willing to participate in pilot coordination
District size	Medium-size — large enough to be representative, manageable in 3 weeks
Existing records	Some partial MSEDCL records available for cross-validation
Security / access	No restricted security zones that would delay early-stage fieldwork

Selection Criterion	Requirement
Connectivity	Good 4G network coverage for real-time data sync during pilot testing

Recommended Pilot District: Nashik District — meets all criteria; accessible from Pune/Mumbai; diverse terrain mix; strong MSEDCL divisional presence. Final selection to be confirmed with MSEDCL management at project kick-off.

Chapter 5: System Architecture & Technology

5.1 System Architecture — Three-Layer Design



5.2 Mobile Survey Application

5.2.1 Core Features

- GPS auto-capture with real-time accuracy indicator — submission blocked if accuracy > 5 metres
- Auto-generation of Unique Pole ID (UID) on record creation — no manual entry needed
- Offline-first architecture — data stored in SQLite locally; syncs to cloud when 2G+ available
- Guided structured form with dropdowns and validation — minimizes free-text errors
- Mandatory 3-photo capture (full pole view, base/foundation, top/equipment)
- Real-time duplicate alert — warns surveyor if within 10m radius of existing submitted pole
- Supervisor QC mode — review, approve, reject, or return records for correction
- Battery and data usage optimization for 8+ hour uninterrupted field use
- Daily productivity counter — shows surveyor their running total against daily target

5.2.2 Data Fields Captured per Pole

Category	Fields Captured
Identity	Unique Pole ID (auto), MSEDCL existing tag (if found), Survey date-time, Surveyor ID
Location	GPS Latitude, GPS Longitude, GPS Altitude, GPS Accuracy (m), Village/Ward, Sub-division, Division, Circle, District
Physical Attributes	Pole material (CC/Steel/Wooden/Tubular), Height range, Condition score (1–5), Tilt observed (Y/N), Visible cracks/damage (Y/N)
Electrical Attributes	Line type (LT/HT/11kV/33kV), Phase (Single/Three), Number of circuits, Transformer mounted (Y/N), Street light attached (Y/N)
Equipment on Pole	DO Fuse, LA (Lightning Arrester), Isolator, LT Distribution Box, Service cable count, Other equipment
Photos	Photo 1 (Full view), Photo 2 (Base/Foundation), Photo 3 (Top/Equipment), Photo 4 (Damage/Special — optional)
Observations	Safety concern flag (Y/N — triggers immediate priority review), Remarks (max 200 characters)

5.3 Backend & Cloud Architecture

Component	Technology	Purpose
Cloud Provider	AWS / Azure / NIC Cloud (TBD by Month 1)	Hosting, compute, storage, CDN
Application Server	Python FastAPI	REST API services, business logic, authentication
Primary Database	PostgreSQL 15 + PostGIS 3	Pole records with full spatial query capability
Photo / File Storage	AWS S3 / Azure Blob Storage	All pole photos (3–4 per pole, estimated 200–300 TB total)
GIS Map Server	GeoServer	Tile rendering, spatial queries, WMS/WFS services
Message Queue	RabbitMQ / AWS SQS	Async sync from 300+ mobile devices simultaneously
Cache Layer	Redis	Dashboard performance — fast map tile serving
Authentication	OAuth 2.0 + JWT tokens	Role-based access control (MSEDCL hierarchy)
API Gateway	AWS API Gateway / NGINX	Rate limiting, security, routing
Monitoring	Grafana + Prometheus	System health, uptime alerts, data pipeline monitoring
Backup	Daily automated to secondary region	Disaster recovery — 30-day point-in-time restore
Mobile Sync CDN	CloudFront / Azure CDN	Fast offline map tile downloads for field devices

5.4 AI Dashboard — MSEDCL's Primary Intelligence Interface

The AI Dashboard is the flagship deliverable of the Intelligence Layer — a fully web-based, role-accessible interface where MSEDCL management can view, analyse, and act on the complete pole network.

Dashboard Modules

- Live Pole Map — Interactive GIS map of all surveyed poles across Maharashtra; filter by district, circle, division, pole type, condition, risk level
- District / Circle Progress Panel — Real-time survey completion percentage per district with RAG (Red/Amber/Green) status
- Condition Heat Map — Geographic visualization of pole condition distribution — identify high-risk zones instantly
- Priority Alert Board — Top poles flagged as Critical (Score 5) or Poor (Score 4) requiring immediate maintenance attention
- AI Classification Dashboard — Breakdown of pole types and condition scores auto-classified by AI models
- Predictive Maintenance Calendar — AI-generated maintenance schedule showing poles due for inspection by month
- Analytics & Trends — Charts showing survey progress over time, condition distribution, district comparisons
- Custom Report Generator — Export pole reports by district/circle/division/condition/type in CSV, Excel, PDF
- Drone Survey Status — Overlay of drone-covered zones vs. ground-surveyed zones
- User Management — MSEDCL role-based access: Corporate view, Circle SE view, Division EE view

5.5 Software Licenses & Tools

Software / Tool	Category	License Type	Cost
QGIS	GIS Desktop Analysis	Open Source	Free
PostgreSQL + PostGIS	Spatial Database	Open Source	Free
GeoServer	GIS Map Server	Open Source	Free
Mapbox GL JS / OpenLayers	Map Visualization (Dashboard)	Free tier / Commercial	₹25,000–60,000/year
AWS / Azure Cloud	Cloud Hosting & Storage	Pay-as-you-use	To Be Finalized
React Native / Flutter	Mobile App Framework	Open Source	Free (dev cost only)
React.js / Vue.js	Dashboard Frontend	Open Source	Free (dev cost only)
TensorFlow 2.x / PyTorch	AI/ML Framework	Open Source	Free

Software / Tool	Category	License Type	Cost
Label Studio	AI Data Labeling Tool	Open Source	Free
MLflow	AI Experiment Tracking	Open Source	Free
Grafana + Prometheus	Monitoring & Alerting	Open Source	Free
DJI Ground Station Pro	Drone Mission Planning	Commercial per device	~₹15,000/device
Pix4D / DroneDeploy	Drone Data Processing	Commercial Subscription	To Be Finalized
Google Workspace / MS 365	Office & Collaboration	Commercial	To Be Finalized

Chapter 6: Methodology

6.1 Methodology Philosophy — Scan-Tag-Verify-Classify-Integrate

6.2 Mobile Survey Workflow

Step 1: Pre-Survey Preparation (Every Morning)

9. Team leader downloads offline zone map for assigned area (village/ward boundary)
10. App displays today's target zone on map — surveyor must stay within assigned boundary
11. Check sync status — all yesterday's records must show green (synced) before departing
12. Confirm device battery >90%, power bank fully charged, SIM data active

Step 2: Active Survey (08:00 – 16:30)

13. Surveyor physically rides / walks along distribution lines in assigned zone
14. On spotting a pole, open 'New Survey' in app — GPS auto-captures (waits for ≤5m accuracy)
15. Fill all mandatory fields using guided dropdown form — takes 2–4 minutes per pole
16. Capture 3 mandatory photos (full view, base, top) + 1 optional damage photo
17. UID auto-generated and displayed — physically write UID on pole with permanent marker where no tag is placed
18. App checks for nearby duplicates — alerts if another pole already recorded within 10m

Step 3: Sync & End-of-Day

19. App syncs every 15 minutes when 2G+ available; force-sync at end of day
20. End-of-day report submitted in app: poles surveyed, issues encountered, tomorrow's zone request
21. Charge phone to 100% overnight — ready for next day

6.3 Drone Survey Workflow

When Drone is Deployed

- Dense forest / tribal areas: Gadchiroli, Chandrapur, Palghar, Nandurbar — drone is primary tool
- Western Ghats steep terrain: Konkan, E. Nashik — drone supplements limited ground access
- Monthly validation sweeps: Random 10% of completed survey zones cross-checked from air

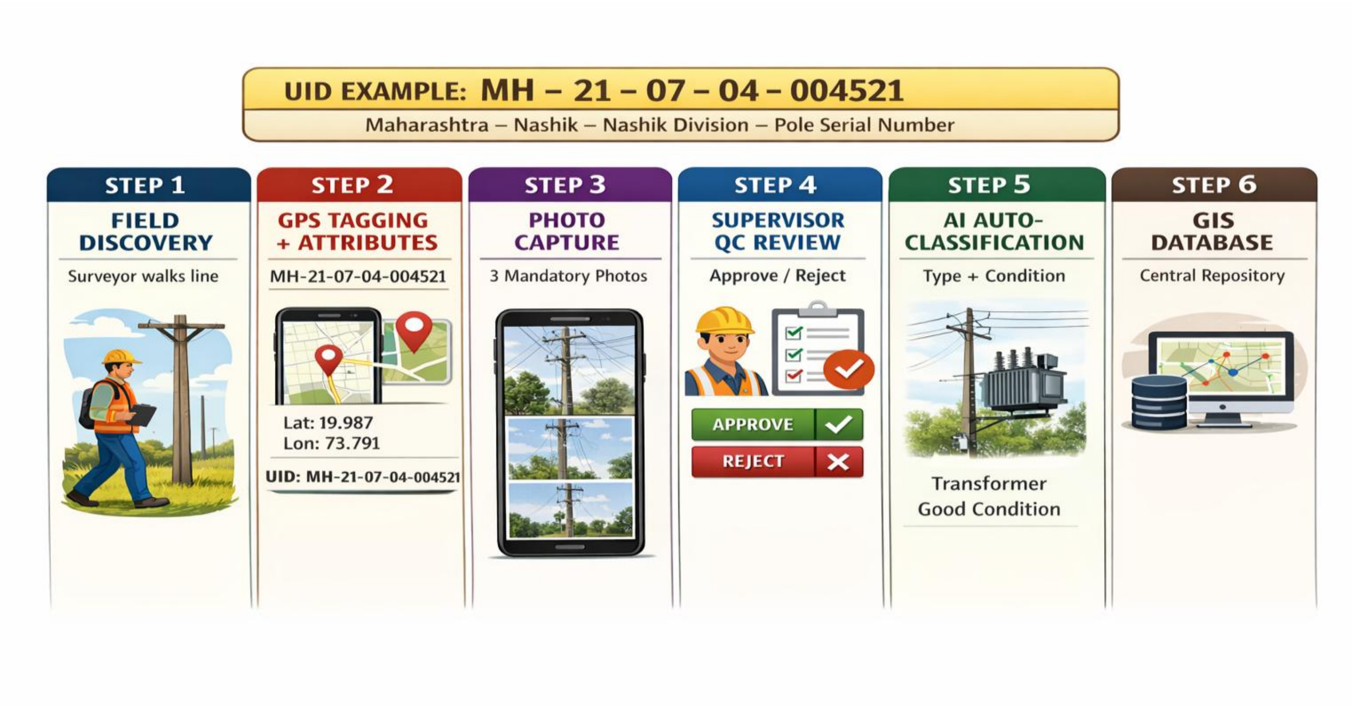
Drone Workflow

22. Mission planning: Define automated flight path in DJI Ground Station / Pix4D Capture — 70% side overlap, 80% front overlap at 60–80m altitude
23. DGCA NPNT compliance: Flight plan submitted via DigitalSky portal before every flight
24. Automated flight execution: Drone captures orthophotos of entire assigned zone
25. Data download and processing: Pix4D / DroneDeploy generates high-resolution orthomosaic map
26. AI pole detection: Computer vision model detects pole locations from orthomosaic image
27. Cross-reference: AI-detected poles compared against mobile survey database for the same zone
28. Gap identification: Poles found by drone but absent in mobile data flagged for ground team revisit

6.4 Quality Assurance (QA) Framework

QA Level	By Whom	Method	When
L1 — Surveyor Self-Check	Surveyor	App validation: mandatory fields, GPS accuracy, photo count — blocks submission if incomplete	Every single record
L2 — Supervisor Field QC	Field Supervisor	Reviews all team's records; physically spot-verifies 10% of submitted poles in the field	Daily
L3 — AI Auto-QC	AI QC Model (Model 4)	Automated photo quality check, GPS outlier detection, duplicate detection — runs on each sync	Real-time on sync
L4 — Central Data QC	Data Management Team	Statistical sampling; cross-district consistency checks; weekly data quality report	Weekly
L5 — Drone Validation	Drone Team + AI	Aerial cross-check of completed zones; identify missed poles	Monthly per district

6.5 Unique Pole ID (UID) System



6.6 Data Collection Standards

Standard	Protocol
GPS Coordinate System	WGS 84 (EPSG:4326) — universal standard; compatible with all GIS platforms
GPS Accuracy Threshold	≤ 5 metres acceptable; ≤ 3 metres preferred; >5m blocked from submission
Photo Resolution Minimum	8 Megapixel minimum; Full HD (1920×1080) — standard on all project devices
Photo Format & Size	JPEG; maximum 5MB per photo (compressed by app before sync to save data)
Date / Time Format	ISO 8601 (YYYY-MM-DD HH:MM:SS) with IST timezone — auto-captured by app
Condition Rating Scale	1 = New/Excellent, 2 = Good, 3 = Fair/Watch, 4 = Poor/Priority, 5 = Critical/Urgent
Language	English for all system fields; Marathi or English for free-text remarks
Offline Storage Limit	Max 1,000 unsynced records before mandatory sync — prevents data accumulation risk

Chapter 7: Implementation Plan & Timeline

⚠ **ALL DATES ANCHORED TO: T0 = April 1, 2026 | T6 = September 30, 2026**
Total project window = 6 months (183 days). Zero timeline slack.

7.1 Phase Overview — 6-Month Master Plan

Phase	Period	Duration	Name	Core Objective
Phase 0	Apr 1 – Apr 7	Week 1 (7 days)	Pre-Deployment Setup	System go-live check, core team mobilization, pilot district preparation
Phase 1	Apr 1 – Apr 21	3 Weeks	Pilot Execution	Full survey of 1 district (Nashik); validate workflow; refine SOPs
Phase 2	Apr 8 – Apr 21	Parallel with Pilot	Mass Hiring & Training	Hire + train all 300 surveyors and 60 supervisors in parallel with pilot
Phase 3	Apr 22 – Aug 15	~4 Months	Full Maharashtra Deployment	All 36 districts simultaneously; 300 surveyors active; AI goes live Month 3
Phase 4	Aug 16 – Sep 15	1 Month	Validation & Completion	Gap-fill, drone sweeps, data quality drive, 100% coverage verification
Phase 5	Sep 1 – Sep 30	Final Month	AI Dashboard + Handover	AI dashboard final, MSEDCL training, documentation, formal handover

7.2 Phase 0 — Pre-Deployment Setup (April 1–7, 2026)

Technology Readiness (Days 1–7)

- Mobile survey app final build deployed to all pilot team devices (30 phones)
- Cloud infrastructure confirmed live — database, APIs, storage, GIS server all running
- Admin dashboard and supervisor review panel live and accessible
- Full end-to-end data pipeline tested: field → sync → database → dashboard
- DGCA drone permit applications submitted (Week 1 critical — 4–8 week processing time)

Pilot District Preparation (Days 1–7)

- MSEDCL divisional office in Nashik briefed; cooperation letter issued
- 30 pilot surveyors and 6 supervisors selected (from pre-screened pool)
- Pilot team devices set up with app, SIM, power banks — field kits ready
- Pilot zone boundaries defined in app system by GIS team
- District Coordinator for Nashik assigned and briefed

7.3 Phase 1 — Pilot Execution (April 1–21, 2026)

Week	Dates	Survey Focus	Parallel Activity
Week 1	Apr 1–7	Urban / town areas of Nashik — dense data, tests app under high-volume conditions	Mass hiring interviews running simultaneously
Week 2	Apr 8–14	Semi-rural and agricultural zones — tests rural workflow and GPS in flat terrain	Batch 1 of 150 surveyors training begins (5 days)
Week 3	Apr 15–21	Remote / hilly zones — tests offline mode, difficult terrain GPS	Batch 2 of 150 surveyors training; SOP updates from pilot feedback

Pilot Success Criteria

- 100% of Nashik district poles surveyed within 3 weeks
- Data completeness rate ≥ 95% on pilot records
- Average productivity of ≥ 50 poles/surveyor/day achieved
- No critical app failures or data loss incidents
- AI labeling team has ≥ 3,000 labeled images by end of pilot for model training
- Pilot report (SOP updates + per-pole cost actuals) completed by April 22

7.4 Phase 2 — Mass Hiring & Training (April 1–21, parallel)

Activity	Dates	Output
Hiring campaign launch	April 1	Job postings on Naukri, college boards, employment exchanges, WhatsApp networks
Interview & shortlisting	April 1–7	360 candidates shortlisted (300 surveyors + 60 supervisors + 10% buffer)
Training Batch 1 (180 pax)	April 8–12	5-day training; surveyors + supervisors; app, GPS, pole ID, safety, SOP
Training Batch 2 (180 pax)	April 15–19	5-day training; second batch; refreshed SOP from pilot week learnings
Device distribution + setup	April 20–21	All 360 field devices set up, apps installed, SIMs active, field kits ready
District assignment	April 21	All surveyors assigned to their specific district; District Coordinators briefed

7.5 Phase 3 — Full Maharashtra Deployment (April 22 – August 15, 2026)

⚠ **April 22, 2026: ALL 36 DISTRICTS GO LIVE SIMULTANEOUSLY**
300 surveyors deployed. 60 supervisors active. 36 District Coordinators in place.
This is the single largest coordinated field deployment day of the project.

Month	Period	Survey Target	AI Milestone	Key Focus
Month 2	Apr 22 – May 31	15–20% of total poles (all 36 districts started)	AI training data accumulating	Stable deployment; productivity ramp-up; SOP adherence
Month 3	Jun 1 – Jun 30	30–40% cumulative	AI MODELS GO LIVE — Jun 30	Peak productivity; AI classification active; data quality drive
Month 4	Jul 1 – Jul 31	55–65% cumulative	AI model refinement	Heavy districts (forest/tribal) drone ops; QA Level 4 audit
Month 5	Aug 1 – Aug 15	75–80% cumulative	AI dashboard integration	Final push; hard-to-reach areas; gap filling begins

District Coordinator Weekly Reporting Rhythm

- Every Monday 09:00: Weekly target agreed with Field Ops Manager
- Every Friday 17:00: Weekly actuals submitted — poles completed, issues, attendance
- Every Sunday: Central data team runs district-wise completeness report

7.6 Phase 4 — Validation & Completion (August 16 – September 15, 2026)

- Identify all districts below 98% completion threshold — deploy additional surveyors
- Drone validation sweeps: All forest/tribal zones cross-checked from air
- Data Quality Audit: Every district record set reviewed for completeness and accuracy
- Gap-filling ground teams dispatched to areas flagged by drone or QC team
- All supervisor QC reviews completed; all pending approvals cleared
- Database locked for new standard submissions by September 15 (corrections-only mode)

7.7 Phase 5 — AI Dashboard, Handover & Closeout (September 1–30, 2026)

- September 1–15: AI Dashboard final testing and MSEDCL UAT (User Acceptance Testing)
- September 1–15: MSEDCL IT team training — 5 days hands-on system administration
- September 10–20: All documentation finalised — technical docs, user manuals, SOPs
- September 20–25: Formal MSEDCL data acceptance — sign-off on database completeness
- September 25–29: System access transfer — all credentials, source code, vendor contacts handed to MSEDCL
- September 30: Final Project Report submitted + Handover ceremony / sign-off meeting

7.8 Critical Milestones Calendar

Milestone	Date	Owner	Success Criteria
M0 — Project Kick-Off	April 1, 2026	Project Director	All core team assembled; app live; cloud live; pilot district ready
M1 — Pilot Complete	April 21, 2026	Field Ops Manager	Nashik 100% surveyed; SOP finalized; pilot report ready
M2 — Full Deployment Day	April 22, 2026	Project Manager	All 36 districts activated; 300 surveyors in field simultaneously
M3 — AI System Live	June 30, 2026	AI / Tech Lead	AI Models 1 & 2 deployed; auto-classifying all incoming records
M4 — 50% State Coverage	July 15, 2026	Field + Data Teams	18+ districts at ≥90% completion; central DB stable
M5 — Survey Complete	August 15, 2026	Field Ops Manager	All 36 districts at ≥95% coverage; gap-filling underway
M6 — 100% Coverage Verified	September 15, 2026	Data Manager + QC	All districts at ≥98%; drone validation done; database locked
M7 — AI Dashboard Live	September 20, 2026	Tech Lead	All dashboard modules functional; MSEDCL UAT passed
M8 — Final Handover	September 30, 2026	Project Director	Handover docs signed; MSEDCL access live; report submitted

7.9 Parallel Workstreams — Week-by-Week Summary

Workstream	Apr 1–21	Apr 22–Jun 30	Jul 1–Aug 15	Aug 16–Sep 30
Field Survey	Pilot (1 district)	Full 36 districts active	Intensive + difficult terrain	Gap-fill + validation
Hiring & Training	Mass hiring + 2 batches	Onboarding replacements	Minimal — only gap fills	Core team only
Tech / App Dev	Pilot fixes + enhancements	Ongoing improvements	Dashboard build	Final UAT + handover
AI Development	Data labeling starts	Training + deployment	Model refinement	Dashboard integration
Drone Operations	Pilot cross-check	Forest / tribal zones	Validation sweeps	Final aerial QC
QA / Data Quality	Pilot QA	Daily L1–L3 automated	Weekly L4 audit	Full L5 district QA
Reporting	Pilot report	Weekly status reports	Monthly dashboards	Final report + handover

Chapter 8: AI Integration — Models, Automation & AI Dashboard

8.1 AI Strategy for 6-Month Timeline

Given the compressed 6-month project window, the AI development strategy is optimized for parallel execution — AI model development runs concurrently with field survey operations from Day 1, with a hard go-live target of June 30, 2026 (end of Month 3). Data labeling begins immediately from the pilot phase, building training data volume rapidly so models are production-ready when field data volumes reach their peak.

8.2 AI Model 1 — Pole Type Classifier

Parameter	Specification
Objective	Automatically identify pole material type from submitted field photos
Model Architecture	Transfer Learning on EfficientNet-B3 / ResNet-50 (pre-trained on ImageNet)
Input	1–4 photos per pole submission (field-captured via survey app)
Output Classes	CC Concrete Pole / RCC Pole / Steel Lattice / Tubular Steel / Wooden Pole / Unknown
Training Dataset Target	Minimum 10,000 labeled images by May 31; 30,000+ by June 30 for production
Target Accuracy (Production)	≥ 90% on held-out test dataset of 2,000 images
Inference Latency	< 3 seconds per pole record processed
Deployment Mode	Cloud API (FastAPI endpoint) — called automatically on every sync

Parameter	Specification
Confidence Threshold	≥ 70% confidence for auto-label; below 70% → flagged for human review
Go-Live Date	June 30, 2026

8.3 AI Model 2 — Pole Condition Assessor

Score	Condition Label	Visual Indicators Detected by AI	Auto-Action Triggered
1	New / Excellent	No damage indicators; clean surface; vertical alignment	Routine inspection schedule (5 years)
2	Good	Minor surface weathering; no structural issues; no lean	Routine schedule (3 years)
3	Fair / Watch	Surface rust patches; minor cracks; slight lean < 3°	Added to watchlist; 1-year inspection
4	Poor / Priority	Significant rust; visible cracks; lean 3–7°; base erosion	Maintenance work order within 3 months
5	Critical / Urgent	Structural failure risk; severe lean > 7°; broken/fractured base	IMMEDIATE alert + replacement work order

Parameter	Specification
Model Architecture	Multi-label CNN + Object Detection (YOLOv8 for defect localization)
Training Dataset Target	15,000+ labeled and condition-graded images by June 30
Target Accuracy	≥ 85% on condition scoring validation set
Special Feature	Defect bounding box — highlights crack/rust location on photo in the dashboard
Go-Live Date	June 30, 2026 (simultaneous with Model 1)

8.4 AI Model 3 — Predictive Maintenance Engine

With 3–4 months of incoming survey data before handover, a basic predictive maintenance engine can be built from Month 4 onwards and refined for handover.

Input Feature	Data Source	Available From
Pole condition score	AI Model 2 output	July 2026
Pole type and material	AI Model 1 output	July 2026
Geographic zone / climate	GIS layer — district and terrain type	April 2026

Input Feature	Data Source	Available From
Proximity to coast / forest	Spatial join with environmental GIS layers	April 2026
Pole age (where known)	MSEDCL records (where available)	Variable
Fault history near pole	MSEDCL complaint data (integration required)	Integration Phase

- Model Output: Risk score (0–100) per pole; poles above threshold added to maintenance priority list
- Target: First predictive report generated by August 31, 2026; handed over with model to MSEDCL

8.5 AI Model 4 — Automated Data Quality Checker

This model runs in real-time on every sync event — ensuring the database receives only clean, validated data. It is the fastest to build (rule-based + lightweight ML) and will be live from the first day of Phase 3 (April 22).

- GPS outlier detection — flags pole coordinates outside assigned district boundary
- Duplicate proximity detection — flags submissions within 5m of an existing record
- Photo blur detection — rejects photos below minimum sharpness threshold
- Photo relevance check — rejects photos that don't appear to show a pole
- Field completeness enforcement — blocks database entry for records missing mandatory fields
- Surveyor anomaly detection — alerts if productivity deviates >50% from team average (above or below)

8.6 AI Development Timeline — Week by Week

Period	AI Development Activity
Apr 1–7 (Week 1)	GPU cloud instances set up; MLflow experiment tracking configured; Label Studio deployed; labeling guidelines written
Apr 1–21 (Pilot phase)	Pilot photos labeled immediately — target 3,000 labeled images; Model 4 (QC checker) built and deployed for April 22 launch
Apr 22 – May 31	Labeling team at full speed — target 15,000 labeled images; Model 1 first training run; Model 2 architecture designed
Jun 1–20	Model 1 and Model 2 trained on full dataset; internal validation and accuracy testing; bug fixes and retraining
Jun 21–30	Model 1 and Model 2 production deployment — integrated into data pipeline; accuracy validation on live data
Jul 1 – Aug 15	Continuous retraining with growing labeled data (target 50,000+ by Aug 15); Model 3 architecture design and initial training
Aug 16 – Sep 15	AI Dashboard integration — all model outputs displayed in dashboard; predictive maintenance module finalized
Sep 16–30	MSEDCL UAT on AI Dashboard; MSEDCL IT team training on AI system; documentation and handover

8.7 Data Labeling Strategy

Parameter	Plan
Labeling Team Size	4 full-time dedicated annotators (hired from April 1)
Labeling Tool	Label Studio — open source, self-hosted on cloud VM, no license cost
Daily Labeling Target	500–600 images per day (team of 4) = ~3,000/week
Total Labeled Images Target	3,000 by Apr 21 → 15,000 by May 31 → 50,000 by Aug 15
Labeling Taxonomy	Pole type (6 classes) + Condition score (5 levels) + Defect tags (rust / crack / lean / damage)
Quality Control	10% of each batch reviewed by AI Lead; inter-rater agreement target ≥ 85%
Labeling Guidelines Doc	Prepared in Week 1; updated after pilot with real field photo examples
Active Learning	From Month 3: AI selects hardest-to-classify images for human labeling — maximizes model improvement per label

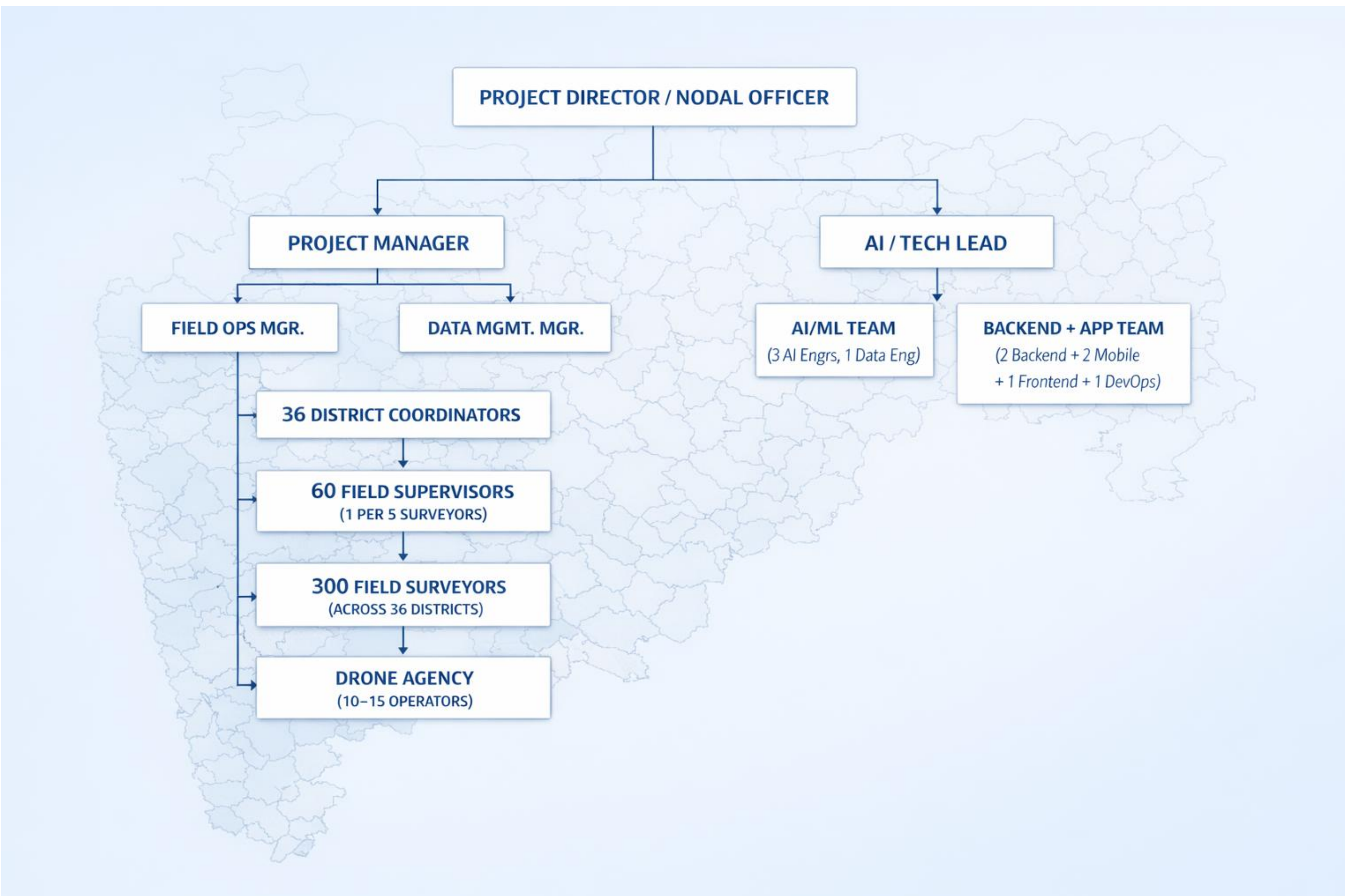
8.8 AI Infrastructure Requirements

Component	Specification	Purpose
GPU Training Instance	AWS p3.2xlarge or equiv. — NVIDIA V100 GPU, 61GB RAM	Model training runs
GPU Inference Instance	AWS g4dn.xlarge — NVIDIA T4 GPU, 16GB RAM	Real-time classification API
Label Studio VM	4 vCPU, 8GB RAM, 500GB SSD	Labeling tool for 4 annotators
MLflow Tracking Server	2 vCPU, 4GB RAM	Experiment tracking, model registry
Model Storage	S3 bucket dedicated for model artifacts	Model version storage
Inference API	FastAPI on Docker + Kubernetes	Production model serving, auto-scaling
AI Framework	TensorFlow 2.x + PyTorch	Model development and training
Model Architecture Base	EfficientNet-B3 (Type), YOLOv8 (Condition/Defect)	Transfer learning on pre-trained weights

Chapter 9: Manpower Planning

⚠ CRITICAL MANPOWER NOTE: Hiring begins April 1, 2026 (T0). There is NO pre-hired team.
Mass hiring and training of ~360 field staff must complete within the first 3 weeks (April 1–21) in parallel with the pilot survey. This is the most time-critical HR activity of the entire project.

9.1 Organisation Structure



9.2 Complete Role Directory

9.2.1 Core Management & Leadership

Role	Count	Qualification	Min. Experience	Key Responsibilities	Start Date
Project Director / Nodal Officer	1	BE/BTech + MBA or senior utility professional	15+ years infra / utility projects	Overall project authority; MSEDCL board interface; final decisions	April 1
Project Manager	1	BE/BTech / MBA Project Management	8–10 years PM experience	End-to-end delivery; milestone tracking; resource management; escalation	April 1
Field Operations Manager	1	Graduate + proven field leadership	7+ years large-scale field ops	Manage all 36 district field teams; logistics; daily operations control	April 1
Data Management Manager	1	BTech/MCA/MSc IT or CS	5+ years data management	Database integrity; data quality oversight; reporting pipeline	April 1
AI / Technology Lead	1	BTech/MTech CS with AI/ML specialization	5+ years AI/ML, 2+ years lead role	AI strategy; system architecture; tech team leadership; MSEDCL tech interface	April 1
GIS Specialist	1	BTech/MSc Geography / GIS / Geomatics	4+ years GIS platforms	GIS platform management; spatial analysis; map layers; dashboard GIS layer	April 1

9.2.2 Technology Team

Role	Count	Qualification	Min. Experience	Key Responsibilities	Start Date
Senior Backend Developer	2	BTech CS / MCA	4+ years Python/Node.js, REST API, PostgreSQL	API development; database schema; cloud infrastructure; pipeline	April 1
Mobile App Developer (Android)	2	BTech CS	3+ years React Native or Flutter, offline-first apps	Survey app development; GPS/camera features; offline sync; QR scanning	April 1
AI / ML Engineer	3	BTech/MTech CS with ML/CV specialization	3+ years ML, computer vision, TensorFlow/PyTorch	Model development; training pipelines; inference deployment; monitoring	April 1

Role	Count	Qualification	Min. Experience	Key Responsibilities	Start Date
Data Engineer	1	BTech CS / MCA	3+ years ETL, data pipelines, database optimization	ETL pipeline; data quality automation; database performance; export tools	April 1
Frontend / Dashboard Developer	1	BTech CS	3+ years React/Vue.js, GIS frontend (Leaflet/Mapbox)	AI dashboard development; GIS map frontend; charts and analytics UI	April 1
DevOps / Cloud Engineer	1	BTech CS	3+ years AWS/Azure, Docker, Kubernetes, CI/CD	Infrastructure-as-code; CI/CD; monitoring setup; security; auto-scaling	April 1
Data Labeling Annotators	4	Any graduate; eye for detail; trainable	Training provided in Week 1	Label training images for AI models; maintain labeling quality standards	April 1
QA / Testing Engineer	1	BTech CS / BCA	2+ years software testing (API + mobile)	Test app, APIs, data pipeline, AI outputs; regression testing; bug reporting	April 1

9.2.3 Field Team

Role	Count	Qualification	Training Required	Key Responsibilities	Start Date
District Coordinator	36 (1 per district)	Graduate; MS Office; Marathi + Hindi / English	2-day orientation on app, reporting format, MSEDCL liaison protocol	Coordinate with local MSEDCL office; manage district surveyors; daily reporting to Field Ops Manager	April 8
Field Supervisor / Team Leader	~60 (1 per 5 surveyors)	Graduate; smartphone literacy; local area knowledge	5-day training: app + QC review process + safety + SOP	Supervise 5 surveyors; perform 10% physical QC daily; approve/reject submissions; productivity tracking	April 12–19
Field Surveyor	~300	10th/12th pass minimum; smartphone literacy essential; physical fitness	5-day intensive: app usage, GPS accuracy, pole identification, condition scoring, safety	Physical pole survey; GPS capture; mandatory 3-photo capture; data entry; UID marking on pole	April 22
Drone Operator	10–15 (via agency)	DGCA Remote Pilot License (RPLC) mandatory; commercial survey experience	Project-specific briefing on zone priority and cross-referencing with mobile data	Drone flights for forest/tribal/difficult zones; orthomosaic generation; pole detection support	May 1

9.2.4 Support Functions

Role	Count	Qualification	Key Responsibilities	Start Date
HR / Recruitment Coordinator	1	MBA HR / Graduate	Mass hiring coordination; onboarding 360 field staff; attendance; replacement hiring throughout project	April 1
Finance / Accounts Executive	1	BCom / MBA Finance	Budget tracking; vendor payments; travel reimbursements; monthly cost reporting	April 1
Legal / Compliance Officer	1	LLB / Graduate	Drone permits; MSEDCL MoU; district permission letters; NDAs for all staff; compliance tracking	April 1
Administrative Support	2	Graduate	Office management; logistics; device procurement; stationery and consumables	April 1
IT Helpdesk / Support	2	BCA / Diploma IT	Device setup; app troubleshooting; remote field tech support; SIM management	April 1

9.3 Hiring Timeline — Critical Path

Activity	Date	Output	Owner
Job postings live (all roles)	April 1, 2026	Postings on Naukri, LinkedIn, college boards, WhatsApp job groups, employment exchanges	HR Coordinator
Core management team assembled	April 1, 2026	Project Director, PM, Field Ops Mgr, AI/Tech Lead, Data Manager, GIS Specialist on board	Project Director
Tech team assembled	April 1, 2026	All 15 tech team members on board; devices set up; coding environments ready	AI/Tech Lead
Support staff on board	April 1, 2026	HR, Finance, Legal, Admin, IT Support (7 people)	Project Manager
District Coordinator selection (36)	April 1–7	36 coordinators selected — preference for local candidates with district knowledge	HR + Field Ops Manager
Supervisor + Surveyor interviews	April 1–7	360 candidates shortlisted — 60 supervisors + 300 surveyors	HR Coordinator + Field Ops Mgr
Labeling team on board (4)	April 1	4 annotators start labeling pilot photos from Day 1	AI Lead
Training Batch 1 (180 pax)	April 8–12	Supervisors for 18 districts + Surveyors Batch 1 trained and certified	Field Ops Manager

Activity	Date	Output	Owner
Training Batch 2 (180 pax)	April 15–19	Remaining supervisors + Surveyors Batch 2 trained; all devices distributed	Field Ops Manager
Full deployment Day	April 22, 2026	ALL 300 surveyors + 60 supervisors active in their districts	Project Manager

9.4 Recruitment Strategy

Surveyor Recruitment (300 — Volume Hiring in 7 Days)

- Primary channel: ITI / Polytechnic colleges in all 5 regions — local candidates strongly preferred
- Secondary channel: District employment exchanges — candidates with basic technical background
- Tertiary channel: Retired MSEDCL linemen / helpers — invaluable field knowledge, fast learners
- WhatsApp job groups per district — spreads quickly through local networks
- Contract-based engagement: 6-month contract aligned with project end date (April 22 – September 30)
- Selection criteria: Smartphone literacy (mandatory), physical fitness, local area familiarity, basic reading/writing

Technical / Management Roles

- Naukri.com and LinkedIn for all technology and management roles
- Engineering college placement cells — Pune, Nagpur, Mumbai, Nashik for fresh talent
- Contract / freelance arrangement for data labelers and junior support roles

9.5 Training Program — 5-Day Module

Day	Module	Duration	Target Audience	Content
Day 1	App & GPS Training	8 hours	All Surveyors + Supervisors	App installation and navigation; GPS accuracy; form filling; UID system; photo guidelines; submission workflow
Day 2	Pole Identification & Scoring	8 hours	All Surveyors + Supervisors	Types of poles (CC/Steel/Wood/Tubular); condition scoring 1–5; equipment identification; common field errors
Day 3	Field Safety & Protocol	8 hours	All Surveyors + Supervisors	Electrical safety distances; never-approach rules near live lines; PPE use; emergency protocol; buddy system
Day 4	Mock Field Survey	Full day	All Surveyors + Supervisors	Supervised practice: 30–40 real poles surveyed under trainer observation; data reviewed live on dashboard
Day 5	QC, Reporting & SOP	8 hours	Supervisors only (extra module); Surveyors do app practice	Supervisor QC workflow; rejection criteria; daily reporting format; escalation protocol; AI output interpretation

9.6 Safety & Personnel Policy

- Group accident insurance: Mandatory for all 360 field staff from Day 1 of deployment
- PPE provided to every surveyor: Reflective safety jacket, helmet, gloves, ID card holder
- Electrical safety rule: Minimum 2-metre distance from any energized line — strictly enforced
- Buddy system: No surveyor works alone in remote areas; minimum 2 people per team
- First aid kit: 1 kit per field vehicle; all supervisors briefed on basic first aid
- Emergency contact tree: Supervisor → District Coordinator → Field Ops Manager — max 30-minute response
- Incident reporting: Any safety incident reported within 1 hour; formal report within 24 hours

Chapter 10: Infrastructure & Logistics

10.1 Infrastructure Overview

Given the 6-month timeline, all infrastructure must be procured and ready by April 1, 2026. There is no ramp-up time for equipment. All devices must be purchased, configured, and distributed before the pilot begins. Vendor contracts for vehicles and drones must be signed in the week before April 1.

10.2 Field Devices & Accessories — Complete List

Device / Item	Qty	Specification	Purpose	Est. Unit Cost (₹)
Android Smartphones	360 (300 surveyors + 60 supervisors)	Min 6.5" display, 64GB, 13MP camera, Android 11+, IP52 rated, 5000mAh battery	Primary field survey tool — app, GPS, photos	12,000–18,000
Spare Smartphones	40 (10% buffer)	Same spec as above — field replacements	Immediate swap for broken/lost devices	12,000–18,000
Power Banks (20,000 mAh)	400 (1 per field staff + spares)	Fast-charge 20,000 mAh; 2 USB-A + 1 USB-C output	Full-day battery backup for 8+ hour field use	1,800–2,500
Ruggedized Phone Cases	400	Shockproof + dustproof military grade; compatible with survey phones	Device protection in harsh field conditions	500–800
Supervisor Laptops	60	i5 Gen 12, 8GB RAM, 256GB SSD, 6+ hr battery, Windows 11	Supervisor QC dashboard; daily reporting; data review	45,000–60,000
Tech Team Laptops	20	i7/Ryzen 7, 16GB RAM, 512GB SSD, developer-grade	AI development; backend; data engineering; dashboards	70,000–90,000
GPU Workstations (AI Dev)	2	NVIDIA RTX 4080/A4000 GPU; 32GB RAM; 1TB NVMe SSD	On-premise AI model development and testing	1,40,000–1,80,000
Dedicated On-Prem Server	1	Dell PowerEdge or HP ProLiant; 2×CPU; 64GB RAM; 4TB RAID storage	Local backup server; dev environment; database mirror	2,50,000–3,50,000

Device / Item	Qty	Specification	Purpose	Est. Unit Cost (₹)
NAS Storage (30TB)	1	QNAP or Synology NAS; RAID 6; 30TB usable storage	Central photo storage backup; large dataset archive	2,00,000–2,80,000
UPS (10 KVA)	1	APC/Eaton 10KVA UPS with 4-hour runtime for server room	Power continuity for HQ critical infrastructure	1,20,000–1,60,000
4G Portable WiFi Routers	70 (1 per supervisor + spares)	Dual SIM 4G; 10-device simultaneous connection; 12hr battery	Team field internet; sync in areas without personal data	3,500–5,000
Label Printers (UID Tags)	5	Brother/Zebra thermal label printer; 20mm width labels	Print QR + UID label stickers for pole tagging	8,000–15,000
UID Label Rolls (QR sticker)	50,000 labels	Weatherproof UV-resistant adhesive; 50mm × 30mm size	Physical UID tagging on poles (accessible poles)	~₹5/label
Safety Kit per Field Staff	360 kits	Reflective vest (ANSI Class 2), hard hat, work gloves, safety boots voucher, ID holder	Field safety compliance — mandatory PPE	1,500–2,500
First Aid Kits	70 (1 per field vehicle)	Standard burn + wound + fracture first aid kit	Field medical emergency preparedness	800–1,200
Measuring Tools	70 sets	5m steel retractable tape + spirit level for lean measurement	Pole height estimation and tilt measurement	400–600
Office Network Infrastructure	1 setup	Gigabit managed switch; WiFi 6 APs (3); firewall; enterprise router	HQ office connectivity for 25-person team	1,80,000–2,50,000
Office Furniture — HQ	1 setup	Desks, ergonomic chairs, storage cabinets, conference table, whiteboards for 25 staff	Core HQ workspace	3,50,000–5,00,000

10.3 Vehicles & Transportation

Vehicle Type	Qty	Purpose	Arrangement	Est. Monthly Cost
Motorcycle / 2-Wheeler	~150 (shared; 1 per 2 surveyors)	Primary movement in rural/narrow-access areas — most efficient for pole-to-pole survey	Surveyor own vehicle + fuel reimbursement; or project-hired for remote districts	₹2,000–3,500 fuel/surveyor
SUV / Bolero 4WD	60 (1 per supervisor team)	Team transport for 5 surveyors + supervisor + equipment; essential for semi-arid and rocky terrain	Monthly vehicle hire with dedicated driver from approved vendor	₹25,000–35,000/vehicle
Mini-Truck / Tempo	5 (shared pool)	Equipment and device transport during district transitions; server/device movement	On-demand hire from logistics vendor	Per-use billing
Drone Operations Vehicle	5 (agency-provided)	Dedicated transport for drone team + equipment; equipment storage overnight	Included in drone agency contract	Covered in agency contract

10.4 Office Infrastructure

10.4.1 Central Project HQ

Item	Specification	Qty	Purpose
HQ Office Space	Min. 2,000 sq. ft., centrally located — Pune or Nagpur preferred	1	Core team of 25 — management, tech, AI, data, support
Meeting / War Room	10-person capacity; 65" display / projector; video conferencing setup	1	MSEDCL presentations; daily management calls; crisis management
Server Room	Minimum 200 sq. ft.; AC; fire suppression; controlled access; UPS connected	1	On-prem server, NAS, UPS, network hardware
Dev Workstations	25 total desks with dual-monitor setup (Dell/HP 24-inch monitors)	25	Tech team + management + data team workspace

10.4.2 District Field Offices (36)

Item	Specification	Qty
Office Space (Rented)	150–300 sq. ft.; near MSEDCL divisional office; ideally same building or adjacent	36 (activated in district waves)
Basic Furniture	2 desks; 4 chairs; 1 whiteboard for daily briefing	36 sets
Internet Connection	Broadband + 4G router backup for daily report sync	36

Item	Specification	Qty
Charging Station	Multi-port USB charging hub (10 ports) for overnight device charging	1 per district
District Notice Board	Printed district zone map; daily targets; SOP checklists	1 per district

10.5 Drone Operations — Agency Selection

Minimum Requirements for Drone Agency

- DGCA-registered operator with valid approvals for commercial survey operations
- Minimum fleet: 5 operational drones — DJI Matrice 300 RTK or Phantom 4 RTK or equivalent
- All pilots must hold valid DGCA Remote Pilot License Certificate (RPLC)
- Proven experience in infrastructure / utility sector drone surveys
- Photogrammetry capability — orthomosaic generation, point cloud processing (Pix4D / DroneDeploy)
- Ability to deploy across multiple districts simultaneously with 48-hour mobilization notice
- SLA: Each drone covers minimum 500 acres per day (at 60m altitude, 70% overlap)

Vendor Tier	Type	Evaluation Criteria	Selection Process
Tier 1 — Primary	Full-service national drone survey companies (ideaForge, Drona Maps, Garuda Aerospace, Asteria)	DGCA certifications, fleet size, Maharashtra experience, SLA, pricing, data processing capability	RFP / Tender — issue by April 1; contract by April 15
Tier 2 — Regional	Maharashtra-based DGCA-registered operators in Pune / Nagpur / Aurangabad	Quick mobilization, local area knowledge, cost efficiency, minimum fleet of 2 drones	Empanelment after Tier 1 contract — regional backup

10.6 Annual Maintenance Contracts (AMC)

Asset	AMC Type	Coverage	Est. Cost
Smartphones — 400 units	Comprehensive with screen + battery replacement	Accidental damage, battery degradation, screen	To Be Finalized
Supervisor Laptops — 60 units	On-site next-business-day service	Hardware failure, battery, keyboard, display	To Be Finalized
Tech Team Laptops — 20 units	On-site next-business-day premium service	Full hardware coverage + loaner device SLA	To Be Finalized
Servers + NAS (On-prem)	24×7 on-site hardware AMC (ProSupport level)	Hardware failure, parts, on-site engineer, 4-hr SLA	To Be Finalized
GPU Workstations — 2 units	Comprehensive hardware AMC	GPU, CPU, RAM, storage failure	To Be Finalized

Asset	AMC Type	Coverage	Est. Cost
Drone Fleet	Included in drone agency SLA contract	Drone crash / malfunction / sensor replacement	Covered in agency contract
Office UPS	Annual preventive maintenance + battery check	Inverter + batteries + load test	To Be Finalized
Cloud Infrastructure	Managed by cloud provider under their SLA	99.99% uptime SLA; no separate AMC needed	Included in cloud cost

10.7 Communication Infrastructure

- WhatsApp Groups (structured hierarchy): All-India Management → Circle-wise → District-wise → Individual team groups
- Project Management Tool: Jira or ClickUp — all tasks, milestones, and issue tracking; mandatory for all roles
- Video Conferencing: Google Meet / MS Teams — daily 15-minute stand-up for management; weekly district coordinator calls
- Official Project Email: All formal communication via project domain email — not personal IDs
- SIM Cards: Each of 400 field devices provided with Jio/Airtel enterprise SIM; 5GB/day data plan
- Emergency Hotline: Dedicated project helpline number for field emergencies — monitored 07:00–20:00 daily

Chapter 11: Budget Framework

11.1 Budget Notes & Assumptions

★ **BUDGET BASIS: Planning-stage estimates for a 6-month project (April 1 – September 30, 2026).**

All human resource / salary costs are marked 'To Be Verified' per MSEDCL HR policy.
Final figures to be confirmed after: (a) MSEDCL scope approval, (b) pilot actuals,
(c) formal vendor RFP/tender responses.
Device costs based on Maharashtra market rates (Q1 2026).
Field operations costs based on 6-month project duration.

11.2 Capital Expenditure (CapEx) — One-Time Procurement

A. Field & Office Devices

Item	Qty	Unit Cost (₹)	Total (₹)	Notes
Android Smartphones (Survey+Supervisor)	400	15,000	60,00,000	Including 40 spares at same spec
Power Banks (20,000 mAh)	400	2,200	8,80,000	1 per field device
Ruggedized Phone Cases	400	650	2,60,000	Military-grade shockproof
Supervisor Laptops	60	52,000	31,20,000	i5, 8GB, 256GB SSD
Tech Team Laptops	20	80,000	16,00,000	i7, 16GB, 512GB SSD
GPU Workstations (AI Dev)	2	1,60,000	3,20,000	RTX 4080 / A4000 GPU
On-Prem Backup Server	1	3,00,000	3,00,000	Dell/HP enterprise server
NAS Storage (30TB)	1	2,40,000	2,40,000	QNAP/Synology RAID 6
UPS 10 KVA	1	1,40,000	1,40,000	APC/Eaton with 4-hr runtime
4G Portable WiFi Routers	70	4,500	3,15,000	Dual SIM, 10 devices
Label Printers (UID tags)	5	12,000	60,000	Thermal, weatherproof labels
UID Label Rolls (weatherproof)	50,000 pcs	5	2,50,000	QR sticker + UID text
Safety Kits (complete per person)	360	2,000	7,20,000	Vest, helmet, gloves, ID holder
First Aid Kits	70	1,000	70,000	One per vehicle
Measuring Tools (tape + level)	70 sets	500	35,000	Pole height estimation

Item	Qty	Unit Cost (₹)	Total (₹)	Notes
Office Network Infrastructure	1 setup	2,20,000	2,20,000	Switch, WiFi AP, firewall, router
Office Furniture (25-person HQ)	1 setup	4,50,000	4,50,000	Desks, chairs, cabinets, boards
SUB-TOTAL — DEVICES & EQUIPMENT			~₹1,49,80,000	Subject to bulk purchase discount

B. Technology Development (CapEx)

Item	Basis	Estimated Cost (₹)
Mobile Survey App (Android)	Custom dev — React Native/Flutter; offline-first; 3-month dev effort	To Be Finalized
Backend API + Cloud Setup	FastAPI + PostgreSQL/PostGIS; initial cloud architecture config	To Be Finalized
GIS Platform Setup	GeoServer setup; base map layers; tile styling; WMS/WFS config	5,00,000–8,00,000
AI/ML Platform Setup	GPU cloud instances; MLflow; Label Studio; training pipeline	To Be Finalized
AI Dashboard Development	Full React frontend; GIS integration; all 10 modules (see Ch.5)	To Be Finalized
System Integration & UAT	End-to-end testing; MSEDCL UAT; bug fixes	To Be Finalized
SUB-TOTAL — TECH DEV (APPROX.)		₹25,00,000–50,00,000 (estimate)

11.3 Operational Expenditure (OpEx) — 6 Months

A. Cloud & Software Costs

Item	Basis	Monthly Cost (₹)	6-Month Total (₹)
Cloud Hosting (App + DB + API)	2–4 production VMs; autoscaling for 300 device syncs	To Be Finalized	To Be Finalized
Cloud Storage (Photos)	Estimated 200–300TB over 6 months; ~3 photos × 3MB × 70 lakh poles	To Be Finalized	To Be Finalized
GPU Cloud (AI Training)	~200 GPU hours/month during training; spot instances used	To Be Finalized	To Be Finalized
SIM Cards + Data Plans (400 devices)	Jio/Airtel enterprise plan; 5GB/day; 400 SIMs	~8,00,000	~48,00,000

Item	Basis	Monthly Cost (₹)	6-Month Total (₹)
GIS / Map Service (Mapbox)	Commercial API plan for dashboard tile serving	~50,000	~3,00,000
Project Management Tool (Jira)	Cloud license for 50 users	~20,000	~1,20,000
Video Conferencing (Teams/Meet)	Enterprise plan for 50 users	~10,000	~60,000
Antivirus / Security Software	Endpoint protection for 80 devices (laptops + workstations)	~15,000	~90,000
SUB-TOTAL (known items)		~8,95,000+	~53,70,000+

B. Field Operations Costs

Item	Basis	Total Cost for 6 Months (₹)
SUV/Bolero Hire (60 vehicles)	₹30,000/vehicle/month × 60 × 5.5 months (Apr 22–Sep 30)	~₹99,00,000
Fuel Reimbursement (2-wheelers)	₹2,500/surveyor/month × 300 × 5 months	~₹37,50,000
District Office Rent (36 offices)	₹10,000/month × 36 offices × avg 5 months	~₹18,00,000
Drone Agency Contract	Per district; per flight hour; orthomosaic processing — subject to RFP	To Be Finalized via RFP
Field Consumables & Printing	UID tags, permanent markers, stationery, printing — per month	~₹3,00,000 (total)
Travel & Accommodation	Inter-district management travel; hotel for training batches	To Be Finalized
Vehicle Driver Cost	Drivers for 60 SUVs — included in vehicle hire rate or separate	Included in hire rate or TBF
SUB-TOTAL FIELD OPS (known)		~₹1,57,50,000+ (excl. drone & travel)

C. Human Resource Costs

Role / Category	Count	Duration (Months)	Monthly Cost	Total Cost
Project Director / Nodal Officer	1	6	To Be Verified	To Be Verified
Project Manager	1	6	To Be Verified	To Be Verified
Field Operations Manager	1	6	To Be Verified	To Be Verified
Data Management Manager	1	6	To Be Verified	To Be Verified
AI / Technology Lead	1	6	To Be Verified	To Be Verified

Role / Category	Count	Duration (Months)	Monthly Cost	Total Cost
GIS Specialist	1	6	To Be Verified	To Be Verified
Backend Developers	2	6	To Be Verified	To Be Verified
Mobile App Developers	2	6	To Be Verified	To Be Verified
AI / ML Engineers	3	6	To Be Verified	To Be Verified
Data Engineer	1	6	To Be Verified	To Be Verified
Frontend / Dashboard Developer	1	6	To Be Verified	To Be Verified
DevOps / Cloud Engineer	1	6	To Be Verified	To Be Verified
Data Labeling Annotators	4	6	To Be Verified	To Be Verified
QA / Testing Engineer	1	6	To Be Verified	To Be Verified
District Coordinators	36	~5 months	To Be Verified	To Be Verified
Field Supervisors	~60	~5.5 months	To Be Verified	To Be Verified
Field Surveyors	~300	5 months (Apr 22 – Sep 30)	To Be Verified	To Be Verified
Drone Operators (via agency)	10–15	Per contract	Included in agency	Per agency contract
Support Staff (7 roles)	7	6	To Be Verified	To Be Verified
TOTAL HR			To Be Verified	To Be Verified

11.4 Budget Summary

Budget Head	Estimated Amount (₹)	Status
A. Devices & Equipment (CapEx)	~₹1,49,80,000	Based on Q1 2026 market rates
B. Technology Development (CapEx)	₹25,00,000–50,00,000	Estimate; finalize after scope confirmation
C. Cloud & Software (OpEx — 6 months)	₹53,70,000+ (known items)	Cloud costs TBF after architecture sizing
D. Field Operations (OpEx — 6 months)	₹1,57,50,000+ (known items)	Drone agency + travel TBF via RFP
E. Human Resources (6 months)	To Be Verified	Per MSEDCL / project HR policy
F. Permissions & Legal	₹10,00,000–15,00,000	Drone permits, MoU, NDAs, district letters
G. Training & Capacity Building	₹10,00,000–15,00,000	Training materials, venues, field kits, refreshments
H. Contingency Reserve (10%)	10% of A+B+C+D	Buffer for unforeseen field / tech expenses
GRAND TOTAL (excl. HR & TBF items)	~₹4,50,00,000–5,50,00,000+	HR costs to be added post HR policy confirmation

Budget Head	Estimated Amount (₹)	Status

⚠ **NOTE: The budget above reflects a 6-month compressed project. All salary/HR figures** require formal HR policy verification. Drone agency costs will be determined via RFP (to be issued April 1). Cloud storage costs will be finalized once data volume estimates are confirmed during the pilot (April 1–21).

Chapter 12: Permissions, Licensing & Legal Framework

⚠ **PERMISSION URGENCY: Given the 6-month timeline, ALL permissions must be initiated**
BEFORE April 1, 2026 where possible. Especially drone permits (DGCA) require 4–8 weeks processing time. Delays in permissions will directly delay field operations.

12.1 MSEDCL Internal Authorizations

Authorization	From	Purpose	Required By
Project Sanction / Work Order	MSEDCL Board / CMD	Official project sanction — foundation for all other approvals	Pre April 1 (completed before T0)
Data Access Authorization	MSEDCL IT / CTO	Access to existing GIS, network maps, SCADA integration	April 1 (T0)
Circle-wise Co-operation Circular	MSEDCL HQ (per circle)	Mandatory directive to all divisional offices to cooperate with survey teams	April 1 (issued before pilot)
Employee / Asset Access Letter	MSEDCL Field Division	Access to MSEDCL premises, substations, infrastructure access	Before each district start
Data Confidentiality Agreement	MSEDCL Legal	Protect MSEDCL's data; define ownership; restrict unauthorized sharing	April 1 (all team members)
MoU / Contract Agreement	MSEDCL Procurement / Legal	Formal contract defining scope, deliverables, timelines, payment	Pre April 1

12.2 Drone / UAV Regulatory Approvals — DGCA

Requirement	Details	Processing Time	Action Required
Unique Identification Number (UIN)	Each drone registered on DGCA DigitalSky portal	2–4 weeks	Drone agency to file before April 1
Remote Pilot License (RPLC)	Every pilot must hold valid DGCA RPLC for Medium category	4–8 weeks	Agency must have certified pilots already
NPNT Compliance	No Permission No Takeoff — all drones must be NPNT hardware/software compliant	Device-level requirement	Confirm with agency at procurement stage
Green Zone Flights	Most rural/agricultural areas — no prior	Real-time per flight	Agency manages via DigitalSky app

Requirement	Details	Processing Time	Action Required
	permission; NPNT compliance sufficient		
Yellow Zone Permission	Within 8–12 km of airports (Nagpur, Nashik, Aurangabad etc.)	48–72 hours per flight plan	Agency submits in advance — part of mission planning
Red Zone Restriction	Near military zones, strategic infra — NO FLY	Not permitted	Survey team avoids; ground survey only
Security Clearance (Gadchiroli / border)	DGCA + MHA clearance for sensitive LWE / border areas	4–12 weeks	Legal officer to file by April 1 — critical path item

12.3 State / District Level Permissions

Permission	From	Purpose	Timeline
District Survey Intimation Letter	District Collector / SP Office	Informs administration of survey activity; prevents team being stopped by police	2 weeks before each district activation
Police Intimation	Superintendent of Police (SP)	Field teams carry SP letter; solves >90% of on-ground questioning by authorities	One-time per district; filed by District Coordinator
Forest Department Permission	Divisional Forest Officer (DFO)	Mandatory for survey in reserved / protected forest areas (Gadchiroli, Chandrapur, Palghar)	Apply by April 1 — 4–6 week processing
Tribal Area Entry Permit (ITR)	District Collector (tribal districts)	Required in Scheduled Area tribal districts for non-tribal entry	Apply early — coordinate through MSEDCL
Revenue Dept. Intimation	Tehsildar (rural areas)	Village-level access for agricultural land poles	Informal coordination + Collector letter sufficient
Cantonment Area Clearance	Local Military Authority	Required in Pune, Nagpur, Aurangabad cantonment areas	Through MoD — apply via MSEDCL formal channel

12.4 Data Privacy & Security Compliance

- Pole infrastructure data is NOT personal data under DPDPA 2023 — no consent needed for pole records
- All cloud storage must be India-based data centers (data residency compliance for government-adjacent project)
- Photo metadata (GPS-tagged) must be stored securely — restricted access only
- All project team members sign NDA before Day 1 — covers data confidentiality and non-disclosure

- Field surveyor location tracking data: Consent obtained at contract signing (DPDPA compliance)
- Data sharing with any third party requires explicit written MSEDCL authorization in advance

12.5 Permission Timeline Tracker

Permission	Filed By	Target Date	Status
MSEDCL Board Project Sanction	Project Director	Pre April 1	To Be Confirmed
MoU / Work Order Execution	Legal Officer + PM	Pre April 1	To Be Confirmed
DGCA Drone UIN (all drones)	Drone Agency	Pre April 1 (critical!)	Pending — Agency to confirm
MSEDCL Circle Co-operation Circular	MSEDCL Nodal Officer	April 1 (Day 1)	To Be Issued
Forest Dept. (Gadchiroli / Chandrapur)	Legal Officer	April 1 (filing)	Pending — 4–6 week lead time
Tribal Area Entry (applicable districts)	Legal + MSEDCL	April 1 (filing)	Pending — Early filing critical
Pilot District — Nashik Collector Letter	District Coordinator (Nashik)	April 1	Pending
All 36 Districts — SP Letters	Legal + 36 Coordinators	Staggered with district activation	Ongoing
Security Clearance (LWE districts)	Legal Officer + MSEDCL	April 1 filing — starts ASAP	Pending — 4–12 weeks
NDA — All Staff (360+)	HR + Legal	April 1 onward (all hires)	Ongoing from T0

Chapter 13: Risk Management

13.1 Risk Context — 6-Month Compressed Timeline

With a 6-month timeline and zero recovery buffer, risk management is not optional — it is a core operational discipline. The project has no room for extended disruptions. Every high-severity risk has a pre-prepared mitigation and a clear escalation owner. Risks are monitored weekly by the Project Manager and reviewed monthly by the Project Director.

13.2 Risk Register

Risk ID	Risk	Category	Severity	Probability	Mitigation Strategy
R01	MSEDCL field staff non-cooperation at divisional level	Operations	CRITICAL	Medium	MSEDCL Board circular before T0; Project Director personally brief Circle SEs; escalation path pre-defined
R02	Mass surveyor hiring failure — cannot onboard 300 in 3 weeks	HR	CRITICAL	Medium	Hiring campaign before April 1; pre-screen candidates; use multiple channels; hire 10% buffer (330 not 300)
R03	Drone DGCA permissions delayed — forest districts uncoverable	Regulatory	HIGH	Medium-High	File ALL drone permits before April 1; start with non-drone areas first; ground survey fallback for delay
R04	Poor 4G connectivity in remote districts — sync failures	Technology	HIGH	High	Offline-first app design; local SQLite storage; surveyors sync at district office daily; no data loss possible
R05	High surveyor attrition in summer heat (May–June)	HR	HIGH	High	Contract with performance bonus at 3-month mark; 30-person replacement bench always maintained; rotate districts
R06	Mobile app critical bugs affecting field productivity	Technology	HIGH	Medium	2-week intensive QA before pilot; pilot as final UAT; hotfix deployment pipeline active from Day 1
R07	Data security breach — pole location data exposed	Security	HIGH	Low	RBAC + JWT auth; encrypted data in transit + at rest; penetration test in Month 1; no public access
R08	AI model accuracy below 85% threshold at June 30 deadline	Technology	HIGH	Medium	Parallel labeling from Day 1; active learning for efficient labeling; human fallback for low-confidence records

Risk ID	Risk	Category	Severity	Probability	Mitigation Strategy
R09	Monsoon disruption to field operations (June–July)	External	HIGH	High	Maharashtra monsoon is predictable — schedule coastal/Konkan area surveys in April–May; Vidarbha for July
R10	Budget overrun due to unforeseen field costs	Financial	HIGH	Medium	Pilot actuals (Apr 1–21) provide real cost baseline; 10% contingency buffer; weekly cost tracking vs. budget
R11	Field safety incident — surveyor injury near live line	Safety	CRITICAL	Low-Medium	Mandatory electrical safety training Day 3; buddy system; no-approach rule; group accident insurance; supervisor spot-checks
R12	Duplicate or poor-quality data entering database at scale	Data Quality	HIGH	Medium	AI QC Model 4 live from April 22; real-time GPS duplicate detection; 10% physical verification by supervisors
R13	Drone agency fails to deliver — missed forest area coverage	Vendor	HIGH	Medium	Two agencies empanelled; minimum 30-day termination notice in contract; activation of Tier 2 within 48 hours
R14	Scope additions by MSEDCL during project mid-flight	Scope	MEDIUM	Medium	MoU defines scope clearly; formal change request process; cost and timeline impact assessment before any approval
R15	Cloud provider outage causing data access disruption	Technology	MEDIUM	Low	On-premise NAS mirror; daily automated export; multi-AZ cloud deployment; 4-hour RTO target

13.3 Critical Risk Response Plans

R01 — MSEDCL Non-Cooperation (CRITICAL)

The single greatest operational risk for this project. An uncooperative divisional office can freeze operations in an entire district. Pre-agreed escalation path: Surveyor issue → District Coordinator → Field Ops Manager (2 hours) → Project Manager (4 hours) → Project Director + MSEDCL Nodal Officer (same day). MSEDCL Board-level directive must name specific consequences for non-compliance. Project Director to do personal calls with all 15 Circle SEs in the week before T0.

R02 — Mass Hiring in 3 Weeks (CRITICAL)

Hiring 300 surveyors and 60 supervisors in 7 days (April 1–7) with training completing April 21 is the tightest single execution challenge in the project. Mitigation: (1) Begin outreach and pre-screening in March 2026 (before T0); (2) Use a structured shortlisting form — screen via WhatsApp first, then 15-minute interview; (3) Hire 10% buffer — 330 surveyors, 66 supervisors; (4) Keep a 'standby bench' of 50 pre-qualified replacements.

R05 + R09 — Monsoon + Summer Attrition

Maharashtra's monsoon begins in Konkan around June 1 and reaches Vidarbha by June 15. This directly overlaps with the peak survey period. Mitigation: (1) Complete all Konkan coastal districts by May 31; (2) Target Vidarbha (Amravati, Akola, Yavatmal) in July when other issues are lower; (3) Provide monsoon gear (rain jacket) to all field staff; (4) Establish daily 'weather hold' decision authority with District Coordinator — do not force field work in unsafe conditions; (5) Summer bonus (small incentive) for completing May target to combat attrition.

R11 — Field Safety Incidents (CRITICAL SEVERITY)

This is non-negotiable. A single fatality or serious injury will halt the project and have severe legal consequences. Requirements: (1) Zero-tolerance approach — any surveyor found violating electrical safety protocol is immediately suspended; (2) First safety stand-down on any incident — all field ops pause for 24 hours for investigation; (3) Group accident insurance covering ₹10 lakh medical + ₹25 lakh accidental death benefit per person from Day 1; (4) Supervisors check PPE compliance every morning before departure.

Chapter 14: Standard Operating Procedures & Monitoring

14.1 Daily Operational Rhythms

Surveyor — Daily SOP

Time	Activity	Output
07:00 AM	Arrive at district base / meeting point; PPE inspection by supervisor	Attendance marked; PPE compliance confirmed
07:15 AM	Morning briefing — today's zone on app; yesterday's issues reviewed; daily target confirmed	Zone downloaded; today's target visible on app
07:30 AM	Depart to survey zone on motorcycle/vehicle	Travel to start point
08:00 AM	Survey commences — first pole of the day	Survey records start populating
12:00 PM	Mid-day sync check — confirm app is syncing; productivity update to supervisor via WhatsApp	Sync status: green; mid-day count reported
01:00 PM	Break (30 minutes); charge phone via power bank	Device charging
01:30 PM	Survey resumes	Afternoon survey
04:30 PM	Return to base; force-sync all records; plug in device for charging	All day's records synced to cloud
05:00 PM	End-of-day report in app: poles surveyed, issues, any unresolved records	Daily count submitted
05:15 PM	Team debrief with supervisor; tomorrow's zone discussed	Next day's plan ready

Supervisor — Daily SOP

Time	Activity
06:30 AM	Review previous day's team records on supervisor dashboard; note any rejected/pending records
07:00 AM	Conduct morning PPE + briefing for team of 5 surveyors; confirm zone assignments
09:00 AM	Physical spot-check: visit 2 surveyors in field; verify GPS accuracy and data quality on 5–10 poles
12:00 PM	Mid-day check-in call with all 5 surveyors; identify anyone behind target; dispatch support if needed
12:30 PM	Submit mid-day progress to District Coordinator: team count, issues, any emergencies
02:00 PM	Review and approve/reject morning's submissions on supervisor app

Time	Activity
04:00 PM	Afternoon field spot-check; look for quality issues in difficult terrain zones
05:00 PM	All submissions reviewed; rejections sent back with clear correction instruction
05:30 PM	End-of-day report to District Coordinator: total approved poles, productivity score, issues, tomorrow's plan

District Coordinator — Daily SOP

- 07:00 AM: Review all supervisors' end-of-previous-day reports; note district productivity vs. target
- 08:00 AM: Submit district morning status to Field Ops Manager on WhatsApp group
- 09:00 AM: Coordinate with local MSEDCL divisional office on any access or cooperation issues
- 12:00 PM: Mid-day productivity check — identify teams below 50% of daily target; escalate if needed
- 04:00 PM: Confirm all supervisors completed their mid-day reports
- 06:00 PM: Submit district daily summary: total poles surveyed today, cumulative, issues, tomorrow's plan

14.2 Data Management SOP

Activity	Frequency	Responsible	Action Required
Sync verification	Daily 09:00	Data Ops Team	Flag any device with >18 hours of unsynced records; alert District Coordinator
AI QC pipeline check	Daily 09:00	AI/Data Team	Review AI QC Model 4 flags: GPS outliers, duplicates, photo failures; send corrections
Data completeness report	Daily	Automated + Data Manager	District-wise completeness score; follow up on records with <90% fields filled
Supervisor approval backlog	Daily	Data Manager	Flag districts where >200 records await supervisor approval for >24 hours
Database backup verification	Daily 23:00	DevOps Engineer	Confirm daily automated backup completed; integrity check weekly
Weekly quality report	Every Monday	Data Manager	District-wise accuracy, completeness, productivity stats submitted to PM and MSEDCL Nodal Officer
Monthly QC audit (Level 4)	1st of each month	Data Manager + AI Lead	Sample 500 records per district; manual cross-check; AI accuracy review; retraining trigger if needed
Drone validation cross-check	Post each drone sweep	Drone Team + Data Team	Compare drone-detected pole positions with mobile survey database; generate gap list for ground team

14.3 AI System SOP

Process	Trigger	Action	Owner
Real-time pole classification	Every sync event	Models 1 & 2 classify pole type and condition; result written to database	Automated (AI Engine)
Low confidence flag	Confidence score < 70%	Record flagged in dashboard as 'Needs Human Review'; data QC team reviews within 24 hours	Data QC Team
Model accuracy review	1st of each month	Run full validation set through both models; compare to ground truth labels; calculate accuracy metrics	AI/ML Engineer
Retraining trigger	Accuracy drops below 85%	Initiate model retraining run with latest labeled dataset; test on validation set before promotion	AI Lead + ML Engineer
Model deployment	After retraining passes threshold	Blue-green deployment — zero downtime; old model serves until new passes health check	DevOps + AI Team
Labeling queue top-up	Daily 09:00	Data Manager ensures labeling team has minimum 1,000 unlabeled images in Label Studio queue	Data Manager
Active learning selection	From Month 3 onwards	AI selects lowest-confidence records for human labeling — maximizes model improvement per label	Automated (MLflow integration)
Model version logging	Every deployment	Version, accuracy, training data size, date logged in MLflow Model Registry	AI/ML Engineer

14.4 Monitoring Dashboard — Management View

KPIs Visible in Real-Time

- Total poles surveyed across Maharashtra — live count and % of target
- District-wise progress — completion %, RAG status (Red <50%, Amber 50–85%, Green >85%)
- Daily productivity — avg poles per surveyor per day (today vs. target vs. project average)
- Data quality score — % records meeting all quality criteria (completeness + GPS + photos)
- AI pipeline status — records classified today, pending classification, flagged for human review
- Sync health map — devices with unsynced data >12 hours shown as red markers on map
- Critical poles alert — live count of poles rated Condition 5 (Critical) requiring immediate action
- Milestone tracker — % achievement vs. project timeline with projected completion date

14.5 Escalation Matrix

Issue Level	Example	First Action	Escalate To	Resolution Target
L1 — Minor	Surveyor confused about form field; app UI question	Self-resolve using SOP guide	Supervisor (if >1 hour)	Same day
L2 — Operational	GPS poor in area; surveyor sick; vehicle delayed	Supervisor resolves	District Coordinator (if >4 hours)	24 hours
L3 — District	MSEDCL office uncooperative; drone permission blocked	District Coordinator + Field Ops Mgr	Project Manager	48 hours
L4 — Project Critical	Safety incident; data breach; system failure; mass attrition	Project Manager immediate	Project Director + MSEDCL Nodal Officer	Immediate (<4 hours)

14.6 Points of Contact (POC)

Internal Project Team

Role	Name	Responsibility	Contact Mode
Project Director	To Be Appointed	Final authority; MSEDCL board interface; critical decisions	Phone + Email (24/7 for emergencies)
Project Manager	To Be Appointed	Day-to-day coordination; all teams; milestone tracking	Phone + WhatsApp + Email
Field Operations Manager	To Be Appointed	All 36-district field teams; logistics; crisis management	Phone + WhatsApp (7 AM – 8 PM)
AI / Technology Lead	To Be Appointed	All technology; AI models; system issues; cloud	Email + Slack/Teams
Data Management Manager	To Be Appointed	Data quality; database; reporting; AI pipeline	Email + Slack/Teams
GIS Specialist	To Be Appointed	Map platform; spatial queries; dashboard GIS layer	Email + Slack/Teams
HR / Recruitment Coordinator	To Be Appointed	All hiring; onboarding; attendance; replacement hiring	Phone + WhatsApp
Legal / Compliance Officer	To Be Appointed	Permits; MSEDCL MoU; all legal matters	Phone + Email
District Coordinators (36)	To Be Appointed per District	District-level ops; MSEDCL liaison; team management	Phone + District WhatsApp group

External / MSEDCL Points of Contact

Organisation	Designation	Responsibility	Contact Method
MSEDCL HQ	Nodal Officer (ED/SE level)	Project approvals; escalation; policy decisions	Phone + Formal letter
MSEDCL Circle (each)	Superintending Engineer (SE)	Circle-level access and cooperation	Phone + Email
MSEDCL Division (each)	Executive Engineer (EE)	Daily field access; sub-division cooperation	Phone
DGCA / DigitalSky	Regional ATC / helpdesk	Drone permits; NPNT compliance support	DigitalSky portal + helpline
Drone Agency	Agency Project Coordinator	Drone scheduling; flight logs; data delivery	Phone + Email + WhatsApp
Cloud Provider	Account Manager (AWS/Azure)	Cloud support; SLA issues; billing	Support portal + dedicated account manager
District Collector (each)	Nodal Officer	Survey permissions; police intimation support	Formal letter + phone

14.7 Reporting Structure

Report	Frequency	Prepared By	Submitted To	Key Content
Daily Field Summary	Daily 6 PM	Each District Coordinator	Field Ops Manager	Poles surveyed, daily target vs. actual, issues, attendance
Weekly Progress Report	Every Monday	Project Manager	Project Director + MSEDCL Nodal Officer	All-district KPIs, AI status, financial summary, risks, decisions needed
Monthly Performance Report	1st of each month	PM + Data Manager + AI Lead	MSEDCL Director / Management	Full KPI dashboard, budget vs. actuals, AI model performance, next month plan
Pilot Completion Report	April 22, 2026	PM + AI Lead + Field Ops Mgr	MSEDCL Management	Pilot findings, per-pole cost actuals, SOP changes, full-deployment readiness confirmation
District Completion Report	On each district completion	District Coordinator + Data Manager	MSEDCL Circle SE + PM	District stats, quality scores, issues log, GIS map extract, handover checklist
AI Model Go-Live Report	June 30, 2026	AI/Tech Lead	Project Director + MSEDCL	Model accuracy results, classification statistics, confidence distribution, known limitations
Final Project Report	September 30, 2026	All Leads	MSEDCL CMD / Board	Full state survey results, AI system documentation, GIS deliverables, handover package, lessons learned

Chapter 15: Final Deliverables — September 30, 2026

★ ALL FIVE FINAL DELIVERABLES MUST BE COMPLETE AND HANDED TO MSEDCL BY SEPTEMBER 30, 2026

(1) Live GIS Dashboard (2) Complete Pole Database (3) AI System Operational
(4) AI Dashboard with full features (5) Final Project Report

D1 — Complete GIS Map of All Maharashtra Poles

Attribute	Specification
Format	Web-based GIS map (browser-accessible) + Export formats: Shapefile (.shp), KML, GeoJSON, CSV
Geographic Coverage	All 36 districts of Maharashtra — 100% of surveyed poles plotted
Access Levels	Corporate (full state view) → Circle SE (circle view) → Division EE (division view) → Sub-division
Spatial Reference	WGS 84 (EPSG:4326)
Map Layers	Pole markers by type, Condition heat map, Risk zone overlay, District/Circle/Division boundaries, Drone-surveyed zones
Update Status	Live during project; static after handover (quarterly update plan to be agreed with MSEDCL)
Export Capability	On-demand export by geography, condition, type, date range — admin dashboard
Handover Date	Map live by August 31; final complete state by September 30, 2026

D2 — Central Pole Asset Database

Attribute	Specification
Database Platform	PostgreSQL 15 + PostGIS 3 — hosted on cloud (AWS/Azure/NIC)
Total Records	Estimated 55–75 Lakh pole records (1 record per unique pole)
Fields per Record	30+ attributes: identity, location (GPS), physical, electrical, equipment, AI scores, photos, history
Photo Storage	Min. 3 photos per pole — cloud object storage; linked from DB by photo ID
Access API	Authenticated REST API — MSEDCL systems can query in real-time
Backup Policy	Daily automated backup; 30-day point-in-time recovery; secondary region mirror
Data Export	CSV, Excel, GIS formats on-demand from admin panel

Attribute	Specification
Ownership	All data is MSEDCL property — credentials and ownership transferred September 30
Handover Date	September 30, 2026

D3 — AI System — Operational & Deployed

AI Component	Description	Accuracy Target	Handover Date
Model 1: Pole Type Classifier	Classifies pole material type from photos — CC/Steel/Wooden/Tubular	≥ 90%	June 30, 2026 (go-live); final Sept 30
Model 2: Condition Assessor	Scores pole condition 1–5 from photos; localizes defects on image	≥ 85%	June 30, 2026 (go-live); final Sept 30
Model 3: Predictive Maintenance	Risk score per pole; maintenance priority list for next 12 months	Best-effort (data-dependent)	August 31, 2026
Model 4: Data QC Checker	Real-time automated validation — GPS outliers, duplicates, photo quality	Real-time, rule-based	April 22, 2026 (live from Day 1 of full deploy)
Labeled Training Dataset	50,000+ labeled pole images — owned by MSEDCL; permanent asset	N/A	September 30, 2026
Full AI Documentation	Model cards, training data spec, API documentation, retraining guide	N/A	September 30, 2026

D4 — AI Dashboard — MSEDCL Intelligence Interface

The AI Dashboard is the premium deliverable of the project — a fully web-based, role-accessible intelligence interface that MSEDCL management will use daily post-handover. It combines the GIS map, AI outputs, analytics, and operational tools in a single unified platform.

Dashboard Feature Modules

Module	Features	Available From
1. Live Pole Map	Interactive GIS map; filter by district/circle/division/type/condition/risk; click any pole for full record + photos	August 31, 2026
2. District Progress Panel	Survey completion % per district; RAG status; daily update; target vs. actual bar chart	April 22, 2026 (internal); August 31 (MSEDCL)
3. Condition Heat Map	Geographic distribution of condition scores across Maharashtra; zoom to any zone; identify regional degradation patterns	August 31, 2026

Module	Features	Available From
4. Priority Alert Board	Top 100 / 500 / 1,000 poles with Condition 5 (Critical) or 4 (Poor) — sorted by risk score; one-click work order export	August 31, 2026
5. AI Classification Panel	Breakdown of pole types state-wide; condition score distribution histogram; confidence score distribution	July 31, 2026
6. Predictive Maintenance Calendar	Month-by-month predicted maintenance requirements by district; export as maintenance plan	August 31, 2026
7. Analytics & Trends	Pole count by type/district/circle; condition trends; survey velocity charts; data quality over time	August 31, 2026
8. Custom Report Generator	User-configurable reports — select fields, filters, geography, date range; export to CSV / Excel / PDF	September 15, 2026
9. Drone Survey Overlay	Map layer showing drone-covered zones, detected pole positions, cross-referenced vs. ground survey	September 15, 2026
10. User & Access Management	MSEDCL role hierarchy management; add/remove users; set access levels (Corporate / Circle / Division)	September 30, 2026

Dashboard Tech Specification	Detail
Frontend	React.js + Tailwind CSS — responsive (desktop + tablet)
GIS Layer	Leaflet.js + Mapbox GL JS — fast tile rendering for 70 lakh pole markers
Charts	Recharts / D3.js — interactive analytics
Backend API	FastAPI — all dashboard data served from central API with caching
Authentication	MSEDCL employee ID integration; role-based access control
Performance Target	Map loads full Maharashtra (all poles visible) within 5 seconds on standard broadband
Mobile Accessibility	Responsive design — usable on tablet for field supervisors
Training Provided	3-day MSEDCL management training; 5-day MSEDCL IT admin training; recorded video tutorials
UAT	September 1–15, 2026 — MSEDCL UAT period with feedback integration
Final Handover	September 30, 2026 — credentials and admin access transferred to MSEDCL IT

D5 — Final Project Report

Report Section	Content
Executive Summary	Project overview, key achievements, total poles surveyed, AI system performance summary
Survey Statistics	District-wise pole counts, type distribution, condition distribution, coverage verification
Technology Documentation	Full system architecture; API reference; database schema; deployment guide
AI System Documentation	Model cards (accuracy, training data, limitations); inference API docs; retraining guide
AI Dashboard Documentation	User manual for all 10 dashboard modules; admin guide; user management guide
Financial Summary	Actual vs. budgeted costs; per-pole cost actuals; cost breakdown by category
Lessons Learned	What worked, what was challenging, recommendations for Phase 2 / ongoing operations
Handover Checklist	All credentials, source code repositories, vendor contacts, SLA documents
MSEDCL Recommendations	Suggested next steps for Phase 2 (integration with SCADA/billing/complaint systems)

Submission: September 30, 2026 — 3 bound hard copies + digital PDF to MSEDCL CMD, Director (Technical), and Nodal Officer

15.1 Deliverables Master Tracker

Deliverable	Type	Target Date	Handover To
GIS Map — Live (full Maharashtra)	Web Application + Export Files	September 30, 2026	MSEDCL IT / GIS Team
Central Pole Database	PostgreSQL + REST API + Credentials	September 30, 2026	MSEDCL IT Team
AI Model 1 — Pole Type Classifier	Deployed API + Model Files + Docs	June 30, 2026 (go-live)	MSEDCL IT / AI Team
AI Model 2 — Condition Assessor	Deployed API + Model Files + Docs	June 30, 2026 (go-live)	MSEDCL IT / AI Team
AI Model 3 — Predictive Maintenance	Deployed API + Model Files + Docs	August 31, 2026	MSEDCL IT / AI Team
AI Model 4 — Data QC Checker	Live pipeline component + Docs	April 22, 2026 (go-live)	MSEDCL IT Team
Labeled Training Dataset (50,000+ images)	Label Studio export + S3 bucket access	September 30, 2026	MSEDCL IT / AI Team
AI Dashboard — All 10 Modules	Web Application + Admin + User Guide	September 30, 2026	MSEDCL IT + Management

Deliverable	Type	Target Date	Handover To
Mobile Survey App	Android APK + Source Code	September 30, 2026	MSEDCL IT Team
Unique Pole ID System	Algorithm + Lookup API + Physical tags on poles	Ongoing; complete Sep 30	MSEDCL Operations
Pilot Completion Report	PDF Report	April 22, 2026	MSEDCL Management
Weekly Progress Reports (26 reports)	PDF / Digital	Every Monday throughout project	MSEDCL Nodal Officer
Monthly Performance Reports (6 reports)	PDF / Dashboard	1st of each month	MSEDCL Director
District Completion Reports (36 reports)	PDF per district	Rolling as each district completes	MSEDCL Circle SE
AI Dashboard User Manual	PDF + Video Tutorials	September 20, 2026	MSEDCL Management + IT
Full Technical Documentation	PDF Document Set	September 30, 2026	MSEDCL IT Team
Final Project Report	Bound PDF + Digital (3 copies)	September 30, 2026	MSEDCL CMD + Director + Nodal Officer

Appendices

Appendix A: Glossary of Terms

Term	Full Form / Meaning
MSEDCL	Maharashtra State Electricity Distribution Co. Ltd.
T0	Project Start Date — April 1, 2026
T6	Project End Date — September 30, 2026
GIS	Geographic Information System — digital mapping and spatial data platform
GPS	Global Positioning System — satellite-based location coordinates
UID	Unique Pole Identifier — permanent digital identity code per pole
AI / ML	Artificial Intelligence / Machine Learning
CNN	Convolutional Neural Network — type of AI model for image analysis
DGCA	Directorate General of Civil Aviation — India's aviation regulator
NPNT	No Permission No Takeoff — DGCA's mandatory digital drone clearance system
UAV	Unmanned Aerial Vehicle (Drone)
RPLC	Remote Pilot License Certificate — DGCA certification for drone pilots
API	Application Programming Interface — software communication protocol
LT	Low Tension — electrical lines below 1 kV (last-mile distribution)
HT	High Tension — electrical lines in 1–33 kV range
EHT	Extra High Tension — lines above 66 kV
PostGIS	Spatial extension for PostgreSQL — enables geographic/spatial queries
CapEx	Capital Expenditure — one-time asset investment
OpEx	Operational Expenditure — recurring ongoing cost
SOP	Standard Operating Procedure
POC	Point of Contact
RAG	Red-Amber-Green — traffic light status indicator in project management
MoU	Memorandum of Understanding — formal cooperation agreement
AMC	Annual Maintenance Contract
WGS 84	World Geodetic System 1984 — universal GPS coordinate reference system
RFP	Request for Proposal — formal tender document for vendor selection
UAT	User Acceptance Testing — MSEDCL's formal testing of system before acceptance

Term	Full Form / Meaning
ITI	Industrial Training Institute
DPDPA	Digital Personal Data Protection Act, 2023
LWE	Left Wing Extremism — relevant for Gadchiroli / Chandrapur district security context
SE	Superintending Engineer — MSEDCL Circle-level senior officer
EE	Executive Engineer — MSEDCL Division-level officer
RBAC	Role-Based Access Control — security model for dashboard user permissions

Appendix B: 6-Month Master Timeline — Gantt Summary

Activity	Apr 1–21	Apr 22–May 31	Jun 1–Jun 30	Jul 1–Jul 31	Aug 1–Aug 31	Sep 1–Sep 30
PILOT (1 District)	ACTIVE	—	—	—	—	—
Mass Hiring & Training	ACTIVE	Replacements only	—	—	—	—
Full 36-District Survey	—	ACTIVE (START)	ACTIVE (PEAK)	ACTIVE	ACTIVE (CLOSE)	Gap fill only
App + Backend Development	Fixes + enhancements	Ongoing	Ongoing	Dashboard build	Dashboard UAT	Final fixes
AI Data Labeling	Pilot photos	Full speed	Full speed	Active learning	Wind down	Final dataset
AI Model Training & Deploy	Architecture	First training	MODELS LIVE	Refinement	Dashboard integration	Final
Drone Operations	Pilot cross-check	Forest zones	Forest zones	Validation sweeps	Final validation	Final aerial QC
GIS Map (Live)	—	Internal	Internal	Internal	MSEDCL access	Final state
AI Dashboard Build	—	—	Design	Build	Build + UAT	FINAL HANDOVER
QA + Validation	Pilot QA	Daily auto-QC	Daily auto-QC	L4 Monthly Audit	Full state QA	Final sign-off
Documentation	Pilot report	Weekly reports	AI go-live report	Monthly report	Monthly report	FINAL REPORT

Appendix C: Pole Condition Reference Guide (Field Card)

Score	Label	What to Look For	Action
1	New / Excellent	No damage; perfectly vertical; clean surface; no corrosion; new appearance	Standard schedule
2	Good	Minor surface weathering; slight discoloration; fully structural; no visible lean	3-year inspection
3	Fair / Watch	Surface rust patches; hairline cracks (not through); lean visible but <3°; some paint peeling	1-year inspection; watchlist
4	Poor / Priority	Heavy rust; significant cracks; lean 3–7°; base erosion visible; bolts loose or missing	Work order — 3 months
5	Critical / Urgent	Structural failure imminent; severe lean >7°; broken base; multiple deep cracks; public safety risk	IMMEDIATE — flag and report now

Appendix D: Complete Pole Data Field Reference (30+ Fields)

Field	Type	Mandatory	Valid Values
pole_uid	Text 20-char	YES	Auto-generated: MH-DD-CC-DV-NNNNNN
survey_datetime	DateTime	YES	Auto: YYYY-MM-DD HH:MM:SS IST
surveyor_id	Text	YES	Auto-populated from login
gps_lat	Decimal 8dp	YES	WGS84 latitude e.g. 19.99512345
gps_lon	Decimal 8dp	YES	WGS84 longitude e.g. 73.78234567
gps_accuracy_m	Decimal	YES	≤5.00 (blocked if >5)
district	Dropdown	YES	36 Maharashtra districts
circle	Dropdown	YES	MSEDCL circle (filtered by district)
division	Dropdown	YES	MSEDCL division (filtered by circle)
sub_division	Dropdown	YES	MSEDCL sub-division
village_ward	Text	YES	Village (rural) / Ward name (urban)
pole_material	Dropdown	YES	CC / RCC / Steel Lattice / Tubular Steel / Wooden / Unknown
pole_height_range	Dropdown	YES	<7m / 7–9m / 9–11m / >11m
line_type	Dropdown	YES	LT / HT / 11kV / 33kV / Service / Unknown
phase_type	Dropdown	YES	Single Phase / Three Phase / N/A
circuit_count	Dropdown	YES	1 / 2 / 3 / More
condition_score	Integer 1–5	YES	1=New to 5=Critical
lean_observed	Boolean	YES	Yes / No
transformer_mounted	Boolean	YES	Yes / No

Field	Type	Mandatory	Valid Values
do_fuse_present	Boolean	YES	Yes / No
la_present	Boolean	YES	Yes / No (Lightning Arrester)
isolator_present	Boolean	YES	Yes / No
lt_box_present	Boolean	NO	Yes / No
street_light	Boolean	NO	Yes / No
service_cable_count	Integer	NO	0–20
safety_concern_flag	Boolean	NO	Yes — triggers immediate priority review
remarks	Text 200-char	NO	English or Marathi; free text
msedcl_existing_tag	Text	NO	MSEDCL pole number if found
photo_1	File Ref.	YES	GPS + timestamp JPEG (full view)
photo_2	File Ref.	YES	GPS + timestamp JPEG (base)
photo_3	File Ref.	YES	GPS + timestamp JPEG (top/equipment)
photo_4	File Ref.	NO	Additional damage photo if needed
ai_type_class	System	Auto	AI Model 1 output — on sync
ai_condition_score	System	Auto	AI Model 2 output — on sync
ai_confidence	System	Auto	0–100% confidence score
sync_status	System	Auto	Local / Synced / Supervisor Reviewed / Approved

Appendix E: Technology Stack Reference

Layer	Component	Technology / Tool
Mobile App	Survey App	React Native (Android-first, iOS future)
Mobile App	Offline Storage	SQLite with WAL mode
Mobile App	GPS	Native Android Location API + GLONASS/BeiDou
Mobile App	Camera	Native Android camera with EXIF metadata
Backend	API	Python FastAPI with async support
Backend	Database	PostgreSQL 15 + PostGIS 3
Backend	Cache	Redis (map tiles + frequent queries)
Backend	Queue	RabbitMQ (300-device simultaneous sync management)
Cloud	Primary	AWS / Azure / NIC Cloud (TBD — April 1)
Cloud	Object Storage	S3 / Blob (photos — est. 200–300 TB total)
Cloud	Container	Docker + Kubernetes (EKS/AKS)
GIS	Map Server	GeoServer 2.x

Layer	Component	Technology / Tool
GIS	Map Client	Leaflet.js + Mapbox GL JS
GIS	Desktop	QGIS 3.x (internal analysis)
AI	Framework	TensorFlow 2.x + PyTorch 2.x
AI	Type Model	EfficientNet-B3 (transfer learning)
AI	Condition Model	EfficientNet + YOLOv8 (defect detection)
AI	Tracking	MLflow (experiments + model registry)
AI	Labeling	Label Studio (self-hosted)
AI	Inference API	FastAPI + ONNX Runtime (CPU/GPU)
Dashboard	Frontend	React.js + Tailwind CSS
Dashboard	Charts	Recharts + D3.js
Monitoring	System	Grafana + Prometheus
Auth	System	OAuth 2.0 + JWT (role-based)
DevOps	CI/CD	GitHub Actions / GitLab CI
DevOps	IaC	Terraform (cloud infra as code)

— END OF DOCUMENT —

MSEDCL Electric Pole Geo-Mapping & AI-Powered Infrastructure Intelligence System

Project Execution Blueprint v2.0 | T0: April 1, 2026 → T6: September 30, 2026 | CONFIDENTIAL